

My Learning Draft

Labix

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1 More on Simplicial Homology

1.1 Relation to Singular Homology

Theorem 1.1.1 Let X be a topological space endowed with a Δ -complex structure $(T, |T| \cong X)$. The induced map $H_n^\Delta(X) \rightarrow H_n(X)$ by the map $s : \Delta^n \rightarrow X$ for each $s \in T^n$ is an isomorphism.

Proof Firstly, every n -simplex $s \in T$ induces a canonical continuous map $s : \Delta^n \rightarrow X$. This extends to a homomorphism $\Delta_n(T) \rightarrow C_n(X)$ and so to a chain map and descends to homology.

Denote T^k the Δ -sets of simplices in T of dimension at most k . Define $\Delta_\bullet(T^k, T^{k-1}) = \Delta_\bullet(T^k)/\Delta_\bullet(T^{k-1})$ and accordingly, $H_n(\Delta_\bullet(T^k, T^{k-1}))$. Then the short exact sequence of chain complexes

$$0 \longrightarrow \Delta_\bullet(T^{k-1}) \longrightarrow \Delta_\bullet(T^k) \longrightarrow \Delta_\bullet(T^k, T^{k-1}) \longrightarrow 0$$

Together with the inclusion maps and by naturality in theorem 1.4.3, give the following:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & H_{n+1}(T^k, T^{k-1}) & \longrightarrow & H_n(T^{k-1}) & \longrightarrow & H_n(T^k, T^{k-1}) & \longrightarrow & H_{n-1}(T^{k-1}) & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & H_{n+1}(|T^k|, |T^{k-1}|) & \longrightarrow & H_n(|T^{k-1}|) & \longrightarrow & H_n(|T^k|, |T^{k-1}|) & \longrightarrow & H_{n-1}(|T^{k-1}|) & \longrightarrow & \cdots \end{array}$$

When $k = 0$, $|T^0|$ is a discrete topological space on the set T^0 , and the map $H_n(T^0) \rightarrow H_n(|T^0|)$ is an isomorphism. By induction and the five lemma, the middle map is an isomorphism provided that the first and fourth maps are isomorphisms.

Note that $\Delta_\bullet(T^{k-1})$ is a chain complex in degrees $k-1, k-2$, down to 0. And $\Delta_\bullet(T^k)$ is the same chain complex with an additional term $\mathbb{Z}T_k$ in degree k . It follows that $\Delta_\bullet(T^k, T^{k-1})$ is the chain complex with $\mathbb{Z}T^k$ concentrated in degree k . In particular, we have that

$$H_n(T^k, T^{k-1}) = \begin{cases} \mathbb{Z}T^k & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

By construction of the geometric realization, we have a homeomorphism

$$\frac{|T^k|}{|T^{k-1}|} \cong \frac{\Delta^k \times T^k}{\partial \Delta^k \times T^k} \cong \bigvee_{T_k} \frac{\Delta^k}{\partial \Delta^k}$$

Using lemma 6.2.4 and lemma 6.2.5, we have that

$$H_n(|T^k|, |T^{k-1}|) \cong \widetilde{H}_n(|T^k|/|T^{k-1}|) \cong \begin{cases} \mathbb{Z}T_k & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

with generators corresponding to elements $s \in T^k$ via the relative cycles $s : \Delta^k \rightarrow |T^k|$. It follows that an isomorphism occurs in the first and fourth position of the long exact sequence of relative homology groups.

If $T = T^n$ for some $n \in \mathbb{N}$, we are done.

Continuing the proof, let $z \in Z_n(T)$ be an n -cycle whose image in $H_n(|T|)$ vanishes. Then there exists $\tau \in C_{n+1}(|T|)$ with $\partial\tau = z$. Now since $|T|$ is the geometric realization, we must have every compact subspace of $|T|$ lying in some $|T^k|$. Hence $\tau \in C_{n+1}(|T^k|)$ for some $k > n$. We deduce that z maps to 0 in $H_n(|T^k|)$. By the previous argument, we have that $z = 0 \in H_n(T^k) = H_n(T)$. This shows injectivity of $H_n(T) \rightarrow H_n(|T|)$.

Similarly, let $\sigma \in Z_n(T)$ be an n -cycle. We have seen that $\sigma \in Z_n(|T^k|)$ for some $k > n$ and by the argument above, $[\sigma]$ comes from $H_n(T^k) = H_n(T)$. This shows surjectivity of $H_n(T) \rightarrow H_n(|T|)$ and so we are done. ■

We can finally show that simplicial homology is independent of the choice of Δ -complex structure.

Corollary 1.1.2 The simplicial homology $H_{\bullet}^{\Delta}(X)$ depends on X only and not on the Δ -complex structure.

Proof Indeed the geometric realization of any two Δ -complex structure is isomorphic to the same singular homology group. ■

Corollary 1.1.3 Suppose X has a Δ -complex structure with simplices in dimension $\leq k$ only. Then $H_n(X) = 0$ for all $n > k$.

Proof Direct since singular homology coincides with simplicial homology. ■

1.2 Simplicial Cohomology

Definition 1.2.1 (Simplicial Cohomology Groups)

Let X be a Δ -complex. Let R be a ring. Let M be an R -module. Define the n th simplicial cohomology group of X

$$H_{\Delta}^n(X; M) = H^n(\text{Hom}(\Delta_{\bullet}(X), M))$$

to be the n th cohomology group of the dual of the simplicial complex $\Delta_{\bullet}(X)$ of X .

1.3

Proposition 1.3.1

Let S be a simplicial complex. Let $\sigma \in S$. Let $p \in |\sigma|^{\circ}$. Then we have

$$H_{\text{sing}}^k(|S|, |S| \setminus p) \cong \tilde{H}_{\Delta}^{k-|\sigma|}(\text{lk}(\sigma))$$

2 Polyhedrons and Polytopes

2.1 Convex Geometry

Definition 2.1.1 (Convex Sets)

Let $C \subseteq \mathbb{R}^n$. We say that C is convex if for all $x, y \in C$ and $t \in [0, 1]$, we have $(1 - t)x + ty \in C$.

Definition 2.1.2 (Convex Hull)

Let $S \subseteq \mathbb{R}^n$ be a subset. Define the convex hull of S to be

$$\text{Conv}(S) = \left\{ \sum_{k=1}^n \lambda_k x_k \mid x_k \in S \text{ and } \sum_{k=1}^n \lambda_k = 1 \right\}$$

Proposition 2.1.3

Let $S \subseteq \mathbb{R}^n$ be a subset. Then the following are sets are equal.

- The convex hull $\text{Conv}(S)$.
- $\bigcap_{C \subseteq \mathbb{R}^n \text{ is convex}} C$.
- The smallest convex set containing S .

2.2 Polyhedrons and its Properties

Polytopes exists more generically in Euclidean space. We lose nothing discussing polytopes in $\mathbb{R}^n$ since Euclidean spaces maybe identified with $\mathbb{R}^n$ canonically.

Definition 2.2.1 (Halfspaces)

A halfspace of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ of the form

$$\{x \in \mathbb{R}^n \mid \langle a, x \rangle + b \geq 0\}$$

for some $a, b \in \mathbb{R}^n$.

Definition 2.2.2 (Polyhedrons)

A polyhedron in $\mathbb{R}^n$ is an intersection of finitely many halfspaces.

Definition 2.2.3 (Polytopes)

A polytope in $\mathbb{R}^n$ is a polyhedron that is bounded.

Proposition 2.2.4

Let $P \subseteq \mathbb{R}^n$ be a subset. Then P is a polytope if and only if there exists a finite subset $S \subseteq \mathbb{R}^n$ such that $P = \text{Conv}(S)$.

Definition 2.2.5 (Dimension)

Let $P \subseteq \mathbb{R}^n$ be a polytope. Define the dimension of P to be the dimension of the smallest affine subspace containing P .

Definition 2.2.6 (Affinely Isomorphic)

Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are affinely isomorphic if there exists an affine transformation $f : \mathbb{R}^{n_1} \rightarrow \mathbb{R}^{n_2}$ such that $f|_{P_1}$ is a bijection.

Note that f itself need not be an affine isomorphism (i.e. an isomorphism of affine spaces).

2.3 The Faces of a Polyhedron

Definition 2.3.1 (Supporting Hyperplane)

Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $H \subseteq \mathbb{R}^n$ be a hyperplane. We say that H is a supporting hyperplane for P if the following are true.

- $P \cap H \neq \emptyset$.
- P lies entirely in one of the halfspaces defined by H .

Definition 2.3.2 (Faces)

Let $P \subseteq \mathbb{R}^n$ be a polyhedron. A face of P is a subset of P defined by

$$F = P \cap H$$

where H is a supporting hyperplane for P .

Lemma 2.3.3

Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face. Then F is a polytope.

Definition 2.3.4 (Vertices, Edges and Facets)

Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Let $F \subseteq P$ be a face.

- We say that F is a vertex if $\dim(F) = 0$.
- We say that F is an edge if $\dim(F) = 1$.
- We say that F is a facet if $\dim(F) = \dim(P) - 1$.

Definition 2.3.5 (The Face Poset)

Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Define the face poset $\mathcal{F}(P)$ of P to be the set of faces of P together with inclusion.

Definition 2.3.6 (The Face Vector)

Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d . Define the face vector $(f_0, \dots, f_{d-1})$ of S by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } P$$

for $0 \leq i \leq d$ and by convention, $f_{-1} = f_d = 1$.

Lemma 2.3.7 (Euler's Formula)

Let $P \subseteq \mathbb{R}^n$ be a polytope of dimension d . Then we have

$$\sum_{i=0}^{d-1} (-1)^i f_i = 1 + (-1)^d$$

Definition 2.3.8 (Combinatorially Equivalent)

Let $P_1 \subseteq \mathbb{R}^{n_1}$ and $P_2 \subseteq \mathbb{R}^{n_2}$ be polytopes. We say that P_1 and P_2 are combinatorially equivalent if there exists a inclusion preserving bijection between $\mathcal{F}(P_1)$ and $\mathcal{F}(P_2)$. In this case we write $P_1 \simeq P_2$.

3 Duality in Polytopes

3.1 Polar Sets

Definition 3.1.1 (Polar Sets)

Let $S \subseteq \mathbb{R}^n$ be a subset. Define the polar set of S to be

$$S^* = \{f \in (\mathbb{R}^n)^* \mid f(x) \leq 1 \text{ for all } x \in S\}$$

Proposition 3.1.2

Let $P \subseteq \mathbb{R}^n$ be a polytope. If $0 \in P$, then P^* is a polytope.

Example 3.1.3

The polar set of the cube $C^3 = \{(a, b, c) \in \mathbb{R}^3 \mid -1 \leq a, b, c \leq 1\}$ is the octahedron

$$(C^3)^* = \{(x, y, z) \in (\mathbb{R}^3)^* \mid \pm x + \pm y + \pm z \leq 1\}$$

$$P = (P^*)^*.$$

$$\mathcal{F}(P)^{\text{op}} \cong \mathcal{F}(P^*).$$

3.2 Simplicial Polytopes

Definition 3.2.1 (Simplicial Polytopes)

Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simplicial polytope if all its facets are simplices.

Lemma 3.2.2

Let $P \subseteq \mathbb{R}^n$ be a polytope. Then ∂P is a simplicial complex.

Proposition 3.2.3 (The Dehn-Sommerville Equation)

Let $P \subseteq \mathbb{R}^n$ be a simplicial polytope of dimension d . Then the following are true:

$$\sum_{j=k}^{d-1} (-1)^j \binom{j+1}{k+1} f_j = (-1)^{d-1} f_k$$

for $-1 \leq k \leq d-2$.

In particular, ∂P is a simplicial complex. The notion of face vector coincides, and they enjoy both the Dehn-Sommerville equation and its relation with the h -vector.

3.3 Simple and Simplicial Polytopes

Definition 3.3.1 (Simple Polytopes)

Let $P \subseteq \mathbb{R}^n$ be a polytope. We say that P is a simple polytope if for all vertices $v \in P$, there exists n edges $e_1, \dots, e_n \subseteq P$ such that $v \in \partial e_i$ for all i .

Lemma 3.3.2

Let $P_1 \subseteq \mathbb{R}^n$ and $P_2 \subseteq \mathbb{R}^m$ be simple polytopes. Then the Cartesian product $P_1 \times P_2$ is a simple polytope.

3.4 The Duality Between Simple and Simplicial Polytopes

Proposition 3.4.1

Let $P \subseteq \mathbb{R}^n$. Then P is simple if and only P^* is simplicial.

Duality of d -dimensional faces with $(n - d)$ -dimensional faces.

Proposition 3.4.2

Let $P \subseteq \mathbb{R}^n$ be a polytope. If P is both simple and simplicial, then P belongs to one of the following types of polytopes.

- $n = 2$ (and so P is a polygon).
- P is a simplex.

3.5 Nerve Complex

Definition 3.5.1 (The Nerve Complex)

Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Define the nerve complex of P to be the simplicial complex

$$\mathcal{K}_P = \partial(P^*)$$

Lemma 3.5.2

Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Then $\mathcal{K}_P$ is a triangulation of S^{n-1} .

Lemma 3.5.3

Let $P_1 \subseteq \mathbb{R}^n$ and $P_2 \subseteq \mathbb{R}^m$ be simple polytopes. Then we have

$$\mathcal{K}_{P_1 \times P_2} \cong \mathcal{K}_{P_1} * \mathcal{K}_{P_2}$$

4 Piecewise Linear Topology

4.1 Relation between Polyhedra and Simplicial Complexes

Definition 4.1.1 (Locally Finite Collections)

Let X be a space. Let $P \subseteq \mathcal{P}(X)$ be a collection of subsets. We say that P is locally finite if for each $p \in X$, there exists a neighbourhood $U \subseteq X$ of p such that the set

$$\{V \in P \mid U \cap V \neq \emptyset\}$$

is finite.

Proposition 4.1.2

Let P be a polyhedron. Then P is a locally finite union of simplexes.

Thus if P is a polytope, then P has the underlying structure of a simplicial complex.

4.2 Piecewise Linear Maps

Definition 4.2.1 (Piecewise Linear Maps between Polyhedra)

Let P, Q be polyhedra. Let $f : P \rightarrow Q$ be a function. We say that f is piecewise linear if there exists a locally finite decomposition into polyhedra $P = \bigcup_{\alpha \in I} P_\alpha$ such that for each $\alpha \in I$, the restriction $f|_{P_\alpha}$ is affine linear.

Proposition 4.2.2

Let P, Q be polyhedra. Let $f : P \rightarrow Q$ be a function. Then the following are equivalent.

- f is piecewise linear.
- For each $p \in P$, there exists $L \subseteq P$ such that $f(\lambda p + \mu x) = \lambda f(p) + \mu f(x)$ for all $\lambda, \mu \geq 0$ and $\lambda + \mu = 1$ and $x \in L$.
- $\Gamma(f) = \{(x, f(x)) \mid x \in P\}$ is a polyhedron.
- There exists locally finite decompositions of P and Q into simplexes such that f sends simplexes to simplexes and f restricted to each simplex is affine linear.

Definition 4.2.3 (PL Homeomorphism)

Let P, Q be polyhedra. Let $f : P \rightarrow Q$ be a piecewise linear function. We say that f is a PL homeomorphism if there exists a piecewise linear function $g : Q \rightarrow P$ such that

$$g \circ f = \text{id}_P \quad \text{and} \quad f \circ g = \text{id}_Q$$

Lemma 4.2.4

Let P, Q be polyhedra. Let $f : P \rightarrow Q$ be a piecewise linear function. If f is a homeomorphism, then f is a piecewise linear homeomorphism.

Motivated by the equivalent definitions of piecewise linear maps between polyhedra, we define piecewise linear maps between spaces in general.

Definition 4.2.5 (Piecewise Linear Maps in General)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a function. We say that f is piecewise linear if there exists locally finite triangulations of X and Y such that f restricted to each simplex is affine linear.

Lemma 4.2.6

Let S, T be simplicial complexes. Let $f : V(S) \rightarrow V(T)$ be a simplicial map. Then $|f| : |S| \rightarrow |T|$ is piecewise linear.

Proposition 4.2.7

Let S, T be simplicial complexes. Let $f : |S| \rightarrow |T|$ be a function. Then f is piecewise linear if and only if there exists subdivisions S' of S and T' of T such that $f : |S'| \rightarrow |T'|$ is simplicial.

4.3 Piecewise Linear Manifolds**Proposition 4.3.1**

The set

$$\Gamma = \{f : U \rightarrow V \mid U, V \subseteq \mathbb{R}^n \text{ are open, } f \text{ is a PL homeomorphism}\}$$

defines a pseudogroup of transformations on $\mathbb{R}^n$.

Definition 4.3.2 (Piecewise Linear Manifolds)

A piecewise linear manifold is a manifold whose pseudogroup of transformations is given by Γ .

4.4 Triangulated Manifolds**Definition 4.4.1 (Triangulated Manifolds)**

Let M be a topological manifold. We say that M is a triangulated manifold if M is triangulated as a topological space.

Proposition 4.4.2

Let X be a piecewise linear manifold. Then X is a triangulated manifold.

Proposition 4.4.3

Let M be a triangulated manifold triangulated by the simplicial complex K . Then the following are equivalent.

- M is a PL manifold.
- For every $\emptyset \neq \sigma \in K$, we have that $\text{lk}(\sigma)$ is PL homeomorphic to $|\Delta^d|$ where $d = \dim(M) - |\sigma|$.
- For every $p \in V(K)$, we have that $\text{st}(p)$ is PL homeomorphic to $|\Delta^{\dim(M)}|$.

The Hauptvermutung asks whether any two homeomorphic triangulated spaces are PL homeomorphic (i.e. if S, T are simplicial complexes and $|S|$ is homeomorphic to $|T|$, is there a PL homeomorphism between $|S|$ and $|T|$?). This is shown to be False by Milnor using Reidemeister torsion.

The manifold Hauptvermutung asks whether any two homeomorphic triangulated manifolds are PL homeomorphic. This is also shown to be false using an obstruction in cohomology.

Proposition 4.4.4 Every finite CW complex is homotopy equivalent to a topological manifold.

4.5 Pseudo-Manifolds**Definition 4.5.1 (Pseudo-Manifolds From Simplicial Complexes)**

Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is a pseudo-manifold of dimension n if the following are true.

- All maximal simplices of $\mathcal{K}$ are of dimension n .
- Each $(n-1)$ -simplex of $\mathcal{K}$ is the face of two n -simplices of $\mathcal{K}$.
- For any two n -simplices $\sigma, \sigma' \in \mathcal{K}$, there exists n -simplices $\sigma = \sigma_0, \dots, \sigma_k = \sigma'$ such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$ -simplex for $0 \leq i \leq k-1$.

Definition 4.5.2 (Pseudo-Manifolds in General)

Let X be a space. We say that X is a pseudo-manifold if X is triangulated by a simplicial complex

$\mathcal{K}$ that is a pseudo-manifold.

Lemma 4.5.3

Let X be a pseudo-manifold of dimension n . Then $H_n(X) \cong \mathbb{Z}$ or 0 .

Lemma 4.5.4

Let $\mathcal{K}$ be a triangulation of S^n . Then the following are true.

- $\mathcal{K}$ is a pseudo-manifold.
- If $\mathcal{K}$ is finite, then the generator of $H_n(\mathcal{K})$ is represented by the sum of n -simplices of $\mathcal{K}$.

5 Cones and Fans

5.1 Cones and its Properties

Definition 5.1.1 (Cones)

Let $C \subseteq \mathbb{R}^n$. We say that C is a cone if $x \in C$ implies that $sx \in C$ for all $s \geq 0$.

Definition 5.1.2 (Polyhedral Cones)

A cone in $\mathbb{R}^n$ is a subset $C \subseteq \mathbb{R}^n$ such that if $c_1, \dots, c_n \in C$, then $\sum_{i=1}^n \lambda_i c_i \in C$ for all $\lambda_1, \dots, \lambda_n \geq 0$.

Definition 5.1.3 (Conical Hull)

Let $S \subseteq \mathbb{R}^n$ be finite. Define the conical hull of S to be

$$\text{Cone}(S) = \left\{ \sum_{s \in S} \lambda_s s \mid \lambda_s \geq 0 \right\}$$

Proposition 5.1.4

Let $C \subseteq \mathbb{R}^n$ be a subset. Then the following are equivalent.

- C is a polyhedral cone.
- There exists a finite subset $S \subseteq \mathbb{R}^n$ such that $C = \text{Cone}(S)$.
- C is the intersection of finitely many halfspaces containing 0 in their boundary.

Therefore cones are also polyhedrons.

Definition 5.1.5 (Dimension of a Cone)

Let $C \subseteq \mathbb{R}^n$ be a cone. Define the dimension of C to be the dimension of the smallest affine subspace containing C .

Definition 5.1.6 (Rational Cones)

Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is rational if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that $C = \text{Cone}(S)$.

Definition 5.1.7 (Regular Cones)

Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is regular if there exists a finite set $S \subseteq \mathbb{Z}^n$ such that the following are true.

- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{Z}$ -linearly independent.

Definition 5.1.8 (Simplicial Cones)

Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is simplicial if there exists a finite set $S \subseteq \mathbb{R}^n$ such that the following are true.

- $C = \text{Cone}(S)$.
- Elements of S are $\mathbb{R}$ -linearly independent.

5.2 The Dual Cone

Definition 5.2.1 (The Dual of a Cone)

Let $C \subseteq \mathbb{R}^n$ be a cone. Define the dual of C to be

$$C^\vee = \{x \in \mathbb{R}^n \mid \langle x, c \rangle \geq 0 \text{ for all } c \in C\}$$

Definition 5.2.2 (Associated Hyperplane and Halfspace of a Linear Functional)

Let $\phi \in (\mathbb{R}^n)^*$. Define the associated hyperplane of ϕ to be

$$H_\phi = \{x \in \mathbb{R}^n \mid \phi(x) = 0\}$$

Define the associated halfspace of ϕ to be

$$H_\phi^+ = \{x \in \mathbb{R}^n \mid \phi(x) \geq 0\}$$

Lemma 5.2.3

Let $C \in \mathbb{R}^n$ be a cone. Let H be a hyperplane in $\mathbb{R}^n$. Then H is a supporting hyperplane of C if and only if there exists $\phi \in C^\vee$ such that $H = H_\phi$.

Lemma 5.2.4

Let $C \in \mathbb{R}^n$ be a polyhedral cone. Then the following are true.

- $C^\vee$ is a polyhedral cone.
- $(C^\vee)^\vee = C$.
- $C = \bigcap_{i=1}^n H_{\phi_i}$ for $\phi_i \in (\mathbb{R}^n)^*$ if and only if $C^\vee = \text{Cone}(\phi_1, \dots, \phi_n)$.
- If C is strongly convex, then $C^\vee$ is strongly convex.

Lemma 5.2.5 (Separation Lemma)

Let $C_1, C_2 \subseteq \mathbb{R}^n$ be cones such that $F = C_1 \cap C_2$ is a face of both C_1 and C_2 . Then there exists $\phi \in C_1^\vee \cap (-C_2)^\vee$ such that

$$F = C_1 \cap H_\phi = C_2 \cap H_\phi$$

Definition 5.2.6 (Strongly Convex Cones)

Let $C \subseteq \mathbb{R}^n$ be a cone. We say that C is strongly convex if C does not contain a line.

Lemma 5.2.7

Let $C \subseteq \mathbb{R}^n$ be a cone. Then the following are equivalent.

- C is strongly convex.
- $\{0\}$ is a face of C .
- $C \cap (-C) = \{0\}$.
- $\dim(C^\vee) = n$.

5.3 Fans and its Properties

Definition 5.3.1 (Fans)

Let $n \in \mathbb{N}^+$. A fan in $\mathbb{R}^n$ is a finite set of cones Σ such that the following are true.

- Each $C \in \Sigma$ is a strongly convex rational polyhedral cone.
- For each cone $C \in \Sigma$ and each face F of C , we have $F \in \Sigma$.
- For cones $C_1, C_2 \in \Sigma$, we have $C_1 \cap C_2$ are faces of C_1 and C_2 .

We write $\Sigma(d) = \{C \in \Sigma \mid \dim(C) = d\}$ for the d -dimensional cones in the fan.

Definition 5.3.2 (The Support of a Fan)

Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Define the support of Σ to be

$$|\Sigma| = \bigcup_{C \in \Sigma} C$$

Definition 5.3.3 (Types of Fans)**Definition 5.3.4** (Complete Fans)

Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. We say that Σ is complete if $|\Sigma| = \mathbb{R}^n$.

6 Some Results on the Classification of Polytopes

Let P be a simplicial polytope of dimension 4. Since P is homeomorphic to D^4 , the following classification problems are equivalent.

- Classification of simplicial polytopes up to uniqueness of the face vector.
- Classification of triangulations of the 3-sphere up to uniqueness of the face vector of the simplicial complex.

(More subtle, this is not true for higher dimensions: see <https://math.mit.edu/~rstan/pubs/pubfiles/54.pdf>)

Of course, two polytopes that are not combinatorially equivalent can still have the same face vector. Let (f_0, f_1, f_2, f_3) be the face vector of P . A few is known in this particular case.

- Since P is simplicial, we have the bound $f_i \leq \binom{f_0}{i+1}$ from here
- The Euler characteristic of the sphere tells us that $f_0 - f_1 + f_2 - f_3 = 0$.
- shows that $f_1 \geq 4f_0 - 10$.
- The same paper tells us that $f_2 = 2f_1 - 2f_0$ and $f_3 = f_1 - f_0$ (also this paper:
- For every pair of (f_0, f_1) such that $f_0 \geq 5$ and $\frac{f_0(f_0-1)}{2} \geq f_1 \geq 4f_0 - 10$, there exists a triangulation of the 3-sphere whose face vectors are $(f_0, f_1, 2f_1 - 2f_0, f_1 - f_0)$.

Recalling that every d dimensional polytope must have $d + 1$ vertices, we start enumerating the possible cases with $f_0 = 5$.

- By the above, we must have $f_1 \geq 10$, and there is guaranteed to be a triangulation if $f_1 \leq 10$. So $(5, 10, 10, 5)$ is one such face vector. By the upper bound of the face vector, this is the only possible face vector.

6.1

Theorem 6.1.1

The following are true regarding the classification of polytopes up to combinatorial type.

- The number of polytopes of dimension d with $d + 1$ vertices is 1.
- The number of polytopes of dimension d with $d + 1$ vertices is $[1/4d^2]$.

6.2

Definition 6.2.1 (Polytopal Spheres)

Let $\mathcal{K}$ be a simplicial complex that is the triangulation of a sphere. We say that the triangulation is polytopal if $\mathcal{K}$ is the boundary of a simplicial polytope.

Convex polytopes p.424

$$\text{Polytopal Spheres} \subseteq \text{PL Spheres} \subseteq \text{Triangulated Spheres}$$

Let d be the dimension of the sphere (hence the underlying polytope has dimension $d + 1$). If $d = 2$, all inclusions are equalities. If the simplicial complex of the triangulation has at most $d + 4$ vertices, all inclusions are equalities.

The smallest non-trivial case is then $d = 3$ such that the polytope has dimension 4 and 8 vertices. The triangulations are classified by "David Barnette. The triangulations of the 3-sphere with up to 8 vertices" where there are a total of 39 triangulations, of which 37 are polytopal, and the polytopes are given in "BRANKO GRUJNBAUM AND V. P. SREEDHARAN. An Enumeration of Simplicial 4-Polytopes with 8 Vertices".

Toric topology p,79

7

Pre-req: Commutative Algebra 2, Simplicial Complexes

7.1 Stanley-Reisner Rings

Theorem 7.1.1

Let R be a commutative ring. Let $n \in \mathbb{N}$. There is a bijection

$$\begin{array}{ccc} \text{Abstract Simplicial} & \xleftrightarrow{1:1} & \text{Squarefree monomial} \\ \text{Complexes on } \{1, \dots, n\} & & \text{ideals of } R[x_1, \dots, x_n] \end{array}$$

given by sending a simplicial set S to the ideal $\langle \prod_{j \in J} x_j \mid J \in \mathcal{P}(\{1, \dots, n\}) \setminus S \rangle$.

Definition 7.1.2 (Stanley-Reisner Rings)

Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Define the Stanley-Reisner ring of S to be

$$R[S] = \frac{k[x_1, \dots, x_n]}{\langle \prod_{j \in J} x_j \mid J \in \mathcal{P}(\{1, \dots, n\}) \setminus S \rangle}$$

The ideal $I_S = \langle \prod_{j \in J} x_j \mid J \in \mathcal{P}(\{1, \dots, n\}) \setminus S \rangle$ is called the Stanley-Reisner ideal of S .

Example 7.1.3

Let R be a commutative ring. The Stanley-Reisner ring of the standard n -simplex Δ^n is given by

$$R[\Delta^n] = k[x_1, \dots, x_n]$$

Example 7.1.4

Let R be a commutative ring. The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1, 2, 3\}) \cup \mathcal{P}(\{1, 3, 4\}) \cup \mathcal{P}(\{2, 4\})$ over $\{1, 2, 3, 4\}$ is given by

$$R[S] = \frac{k[x_1, x_2, x_3, x_4]}{(x_1 x_2 x_4, x_2 x_3 x_4)}$$

Proposition 7.1.5

Let R be a commutative ring. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Then we have

$$I_S = \bigcap_{J \in S} \langle x_i \mid i \notin J \rangle$$

Maps of simplicial sets $\rightarrow$ Maps of k -algebra.

Proposition 7.1.6

Let R be a commutative ring. Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Then we have

$$R[\mathcal{K}_1 * \mathcal{K}_2] \cong R[\mathcal{K}_1] \otimes_R R[\mathcal{K}_2]$$

Proposition 7.1.7

Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}$ is a flag complex, then $R[\mathcal{K}]$ is a Koszul algebra.

7.2 The Hilbert Series of Stanley-Reisner Rings

Theorem 7.2.1 (Stanley)

Let k be a field. Let $n \in \mathbb{N}$. Let S be an abstract simplicial complex on $\{1, \dots, n\}$. Let $(f_0, \dots, f_n)$ be the face vector of S . Let $(h_0, \dots, h_n)$ be the h -vector of S . Then the Hilbert series of $k[S]$ is given by

$$HS_{k[S]}(t) = \sum_{i=0}^{n+1} f_{k-1} \left(\frac{t^2}{1-t^2} \right)^k = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} h_i t^{2i}$$

Example 7.2.2 | Let Δ^n be the standard n -simplex. Then the following are true.

- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by

$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$

- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by

$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} t^{2i}$$

8 Moment-Angle Complexes

Pre req: Equivariant (co)homology, homological algebra, PL manifolds?

8.1 Basic Definitions

Proposition 8.1.1

Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \rightarrow \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \rightarrow (|z_1|^2, \dots, |z_n|^2)$. Suppose that $(S^1)^n \subseteq \mathbb{C}^n$ act on $\mathbb{C}^n$ by left multiplication. Then μ descends to a homeomorphism of orbit spaces

$$\frac{\mathbb{C}^n}{(S^1)^n} \cong \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } i\} \subseteq \mathbb{R}^n$$

Lemma 8.1.2

Let $n \in \mathbb{N}$. Let $\mu : \mathbb{C}^n \rightarrow \mathbb{R}^n$ be the map $(z_1, \dots, z_n) \rightarrow (|z_1|^2, \dots, |z_n|^2)$. Then the above homeomorphism induces a homeomorphism

$$\frac{D^n}{(S^1)^n} \cong [0, 1]^n$$

Recall the construction of the cubical embedding $i_c : \Delta^n \rightarrow [0, 1]^{n+1}$.

1. For $1 \leq k \leq n$, the barycenter of the k -dimensional face $B([e_{i_1}, \dots, e_{i_k}])$ is sent to $(u_1, \dots, u_n) \in [0, 1]^{n+1}$ where $u_j = 0$ if $j \in \{i_1, \dots, i_k\}$ and $u_j = 1$ otherwise.
2. Extend this into a piecewise linear embedding $i_c : \Delta^n \rightarrow [0, 1]^{n+1}$ as follows. For each simplex $[p_1, \dots, p_l]$ in the barycentric subdivision of Δ^n , $[i_c(p_1), \dots, i_c(p_l)]$ lies on a face of $[0, 1]^{n+1}$. Hence there exists a simplicial map sending $[p_1, \dots, p_l]$ to $[i_c(p_1), \dots, i_c(p_l)]$.

Since every finite abstract simplicial complex can be realized as a subcomplex of Δ^n for large n , we also get the cubical embedding of a finite abstract simplicial complex, which is denoted by $\text{cc}(\mathbb{K})$.

Definition 8.1.3 (Moment-Angle Complex)

Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Define the moment-angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = \mu^{-1}(\text{cc}(\mathcal{K})) \subseteq D^n$$

Equivalently, it is the pullback of the inclusion map $\mathcal{Z}_{\mathcal{K}} = \text{cc}(\mathcal{K}) \rightarrow [0, 1]^n$ and the quotient map $\mu : D^n \rightarrow [0, 1]^n$.

Example 8.1.4

Let $n \in \mathbb{N}$. Then the following are true.

- $\mathcal{Z}_{\emptyset} = (S^1)^n$.
- $\mathcal{Z}_{\Delta^n} = D^{n+1}$.

Lemma 8.1.5

Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. The following are true regarding the moment-angle complex of $\mathcal{K}$.

- $\mathcal{Z}_{\mathcal{K}}$ is a $(S^1)^n$ -invariant subspace.
- The homeomorphism $\frac{D^n}{(S^1)^n} \cong [0, 1]^n$ induced by the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$ descends to a homeomorphism

$$\frac{\mathcal{Z}_{\mathcal{K}}}{(S^1)^n} \cong \text{cc}(\mathcal{K})$$

Recall that $\text{cc}(\mathcal{K})$ is equal to the union of a finite number of faces of $[0, 1]^n$. Also recall the notation

$$C_I = \{(x_1, \dots, x_n) \in [0, 1]^n \mid x_i = 0 \text{ if } i \in I\}$$

for a face of the cube $[0, 1]^n$ that contains the origin and $I \subseteq \{1, \dots, n\}$.

Lemma 8.1.6

Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex over $\{1, \dots, n\}$. Suppose that $\text{cc}(\mathcal{K}) = \bigcup_{I \in \mathcal{K}} C_I$. Then there is a homeomorphism

$$\mathcal{Z}_{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \mu^{-1}(C_I) = \bigcup_{I \in \mathcal{K}} \{(x_1, \dots, x_n) \in D^n \mid |z_j|^2 = 1 \text{ for all } j \in \{1, \dots, n\} \setminus I\}$$

where μ is the map $\mu(z_1, \dots, z_n) = (|z_1|^2, \dots, |z_n|^2)$.

For $I \subseteq \{1, \dots, n\}$, the condition that $|z_j|^2 = 1$ for $j \in \{1, \dots, n\} \setminus I$ means that for those coordinates in D^n , they only consist of the boundary which is S^1 . Hence we may think of $\mathcal{Z}_{\mathcal{K}}$ as the union of products

$$\mathcal{Z}_{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \left(\prod_{i \in I} D^2 \times \prod_{i \in \{1, \dots, n\} \setminus I} S^1 \right)$$

But note that this product is ambiguous on which coordinates should be S^1 and which should be D^2 (because we have a chosen embedding into $\mathbb{C}^n$ although swapping the order of the products give homeomorphic spaces). Moreover, because we are taking the union over all $I \in \mathcal{K}$, it suffices to consider the maximal simplices in $\mathcal{K}$.

Example 8.1.7

Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are true.

- The moment angle complex of $\partial\Delta^n$ is given by

$$\mathcal{Z}_{\partial\Delta^n} \cong S^{2n+1}$$

- The moment angle complex of the n -gon is given by

Proof

It suffices to consider the maximal simplices in $\partial\Delta^n$. They are of the form $I \subseteq \{1, \dots, n+1\}$ for $|I| = n$. Hence we have

$$\begin{aligned} \mathcal{Z}_{\partial\Delta^n} &= (D^2 \times \dots \times D^2 \times S^1) \cup \dots \cup (S^1 \times D^2 \times \dots \times D^2) \\ &= \partial(D^2 \times \dots \times D^2) \\ &\cong \partial D^{2n+2} \\ &\cong S^{2n+1} \end{aligned}$$

For the moment angle complex of the n -gon, we note that the simplicial complex is given by 1-simplices of the form $\{i, i+1\}$ for $1 \leq i \leq n$ together with $\{1, n+1\}$. Hence

$$\mathcal{Z}_{n\text{-gon}} = (D^2 \times D^2 \times S^1 \times \dots \times S^1) \cup \dots \cup (S^1 \times \dots \times S^1 \times D^2 \times D^2) \cup (D^2 \times S^1 \times \dots \times S^1 \times D^2)$$

■

Proposition 8.1.8

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1} , then $\mathcal{Z}_{\mathcal{K}}$ is a closed topological manifold of dimension $n + m$.

8.2 Polyhedral Products

Definition 8.2.1 (Polyhedral Products)

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A}) = \{(X_1, A_1), \dots, (X_n, A_n)\}$ be pairs of spaces. Define the polyhedral product of $(\mathbf{X}, \mathbf{A})$ corresponding to $\mathcal{K}$ to be

$$(\mathbf{X}, \mathbf{A})^{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} \left\{ (x_1, \dots, x_n) \in \prod_{i=1}^n X_i \mid x_i \in A_i \text{ for } i \notin I \right\} = \bigcup_{I \in \mathcal{K}} \left(\prod_{i \in I} X_i \times \prod_{i \notin I} A_i \right)$$

8.3 Algebraic Invariants of the Moment Angle Complex**Proposition 8.3.1**

Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- There is an isomorphism of rings

$$H^*((\mathbb{CP}^\infty, *)^{\mathcal{K}}; R) \cong R[\mathcal{K}]$$

- The inclusion $(\mathbb{CP}^\infty, *)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty, *)^n$ induces the quotient map

$$H^*((\mathbb{CP}^\infty)^m; R) \cong R[v_1, \dots, v_n] \rightarrow R[\mathcal{K}] \cong H^*((\mathbb{CP}^\infty, *)^{\mathcal{K}}; R)$$

(where $v_1, \dots, v_n$ are degree 2 elements).

- The inclusion $i : (\mathbb{CP}^\infty, *)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty, *)^n$ decomposes into a homotopy equivalence and fibration in the following way:

$$\begin{array}{ccc} (\mathbb{CP}^\infty, *)^{\mathcal{K}} & \xhookrightarrow{\quad} & (\mathbb{CP}^\infty, *)^n \\ & \searrow \cong & \nearrow \text{fib.} \\ & ET^m \times_{T^m} \mathcal{Z}_{\mathcal{K}} & \end{array}$$

Moreover, the fiber of the fibration is $\mathcal{Z}_{\mathcal{K}}$.

- $\mathcal{Z}_{\mathcal{K}}(BS^1, *) = ET^m \times_{T^m} \mathcal{Z}_{\mathcal{K}}$.
- $H^*(ET^m \times_{T^m} \mathcal{Z}_{\mathcal{K}}) = R[\mathcal{K}]$.

Theorem 8.3.2 ((Davis-Januskiewicz))

Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the equivariant cohomology ring of $\mathcal{Z}_{\mathcal{K}}$ is given by

$$H_{T^m}^*(\mathcal{Z}_{\mathcal{K}}; R) \cong R[\mathcal{K}]$$

(natural in $\mathcal{K}$)

Proposition 8.3.3

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true regarding the homotopy groups of $\mathcal{Z}_{\mathcal{K}}$.

- $\mathcal{Z}_{\mathcal{K}}$ is two-connected.
- $\pi_i(\mathcal{Z}_{\mathcal{K}}) \cong \pi_i((\mathbb{CP}^\infty, *)^{\mathcal{K}})$ for $i \geq 3$.

Consider the decomposition of D^2 into the following CW complex structure:

- One 0-cell $1 \in D^2$.
- One 1-cell S^1 .
- One 2-cell D^2 .

Let $n \in \mathbb{N}$. This gives the product complex D^{2n} a CW complex structure, whose cells are a product of 1-cells and 2-cells. We parameterize the cells in the following way. For $J, I \subseteq \{1, \dots, n\}$ with $J \cap I = \emptyset$, we denote $\chi(J, I)$ as the $(|J| + |I|)$ -cell corresponding to the cell with the factor of a 1-cell in the j th position for $j \in J$, the factor of a 0-cell in the i th position for $i \in I$, and the factor of a 0-cell for $k \notin I \cup J$.

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a union of cells in D^{2n} , and so is a cellular subcomplex of D^{2n} . Namely, for each $I \in \mathcal{K}$, we have that $\chi(J, I) \subseteq \mathcal{Z}_{\mathcal{K}}$ for all $J \subseteq \{1, \dots, n\} \setminus I$.

Q: How does the cellular cochain split into a bicomplex

Definition 8.3.4 (Bigrading in the Cochain Complex)

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define a bigrading in $C^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$ as follows. For each basis element $\chi(J, I)$, define $\deg(\chi(J, I)) = (-|J|, 2(|I| + |J|))$.

Lemma 8.3.5

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- The $(2j)$ th components for each $j \in \mathbb{N}$ defines a cochain complex $C^{-i, 2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$ in i .
- $(C^{-*, *}, d_{CW}, 0)$ is a double cochain complex.
- The cellular cochain complex of $\mathcal{Z}_{\mathcal{K}}$ splits into a direct sum

$$C^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \text{Tot}^{\oplus}(C^{-*, *}) = \bigoplus_{j \in \mathbb{N}} C^{-*, 2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$$

where

$$C^k(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \bigoplus_{2j-i=k} C^{-i, 2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$$

The consequence of the splitting is that cohomology also acquires a splitting

$$H^n(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \bigoplus_{2j-i=n} H^{-i}(C^{-*, 2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}))$$

Proposition 8.3.6

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- There are natural isomorphisms

$$H^{-p}(C^{-*, 2q}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})) \cong H_{-p} \left(\Lambda \left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i \right) \right)_{2q} = H_{-p} \left(\Lambda \left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i \right) \right)_{2q}$$

where $\Lambda(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i)$ is the differential graded algebra whose homological grading is given by $\deg(u_i) = -1$, internal grading is given by $\deg(v_i) = 2$ and differential is given by $d(u_i) = v_i$ (and $d(v_i) = 0$ since we think of the DGA as a DGA over the ring $\mathbb{Z}[\mathcal{K}]$).

- There above natural isomorphisms induces natural isomorphisms

$$H^k(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p} \left(\Lambda \left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i \right) \right)_{2q}$$

- The above natural isomorphisms induce a natural isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong H_* \left(\Lambda \left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i \right) \right)$$

Note that $H^n(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$ is not isomorphic to $H_{-n}(\Lambda(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i))$ despite the suggestive notation. This is a mismatch of algebraic objects since the left is an abelian group while the right hand side is a $\mathbb{Z}[\mathcal{K}]$ -module. In order to have an isomorphism of abelian groups we must further decompose each homology $\mathbb{Z}[\mathcal{K}]$ -modules further into $\mathbb{Z}$ -modules by considering $\mathbb{Z}[\mathcal{K}]$ as a $\mathbb{Z}$ -module. But this results in a shift of degrees.

Let $K_{\bullet}(v_1, \dots, v_n; \mathbb{Z}[\mathcal{K}])$ be the Koszul complex over $\mathbb{Z}[v_1, \dots, v_n]$. Notice that the cochain complex defined by $K^n = K_n(v_1, \dots, v_n; \mathbb{Z}[\mathcal{K}])$ is precisely our differential graded algebra $\Lambda(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i)$. Then

the cohomology of $\mathcal{Z}_{\mathcal{K}}$ acquires an additional description:

$$H^n(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \bigoplus_{2q-p=n} (H_p(K_{\bullet}(v_1, \dots, v_n; \mathbb{Z}[\mathcal{K}])))_{2q}$$

Finally, since $K_{\bullet}(v_1, \dots, v_n)$ is a free resolution for $\mathbb{Z}$ as a $\mathbb{Z}[v_1, \dots, v_n]$ -module, we know that

$$H_p(K_{\bullet}(v_1, \dots, v_n; R[\mathcal{K}])) = \text{Tor}_p^{\mathbb{Z}[v_1, \dots, v_n]}(\mathbb{Z}[\mathcal{K}], \mathbb{Z})$$

The Koszul algebra gives the Tor groups the structure of an algebra, and so the cohomology ring acquires the description

$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \text{Tor}_*^{\mathbb{Z}[v_1, \dots, v_n]}(\mathbb{Z}[v_1, \dots, v_n], \mathbb{Z})$$

But again note that the n th cohomology group of $\mathcal{Z}_{\mathcal{K}}$ is not isomorphic to the n th Tor group of $\mathbb{Z}[\mathcal{K}]$ and $\mathbb{Z}$ by virtually the same reason as above.

Proposition 8.3.7

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- There are natural isomorphisms of cochain complexes

$$\bigoplus_{J \subseteq \{1, \dots, n\}} C^{k-|J|-1}(\mathcal{K}_J; \mathbb{Z}) \cong C^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$$

given by

$$\alpha_L \mapsto \varepsilon(L, J) \chi(J \setminus L, L)$$

where α_L is the cochain basis of Hochster's formula given by the simplex $L \subseteq J$ and

$$\varepsilon(L, J) = \prod_{j \in L} \varepsilon(j, J)$$

and $\varepsilon(j, J) = (-1)^{r-1}$ if j is in the r th position of J .

- There above natural isomorphisms induces natural isomorphisms

$$H^{-i}(C^{-*, 2j}(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})) \cong \bigoplus_{\substack{J \subseteq \{1, \dots, n\} \\ |J|=j}} \tilde{H}^{j-i-1}(\mathcal{K}_J; \mathbb{Z})$$

where by convention we set $\tilde{H}^{-1}(\mathcal{K}_{\emptyset}) = \mathbb{Z}$.

- The above natural isomorphisms induces natural isomorphisms

$$H^k(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \bigoplus_{J \subseteq \{1, \dots, n\}} \tilde{H}^{k-|J|-1}(\mathcal{K}_J; \mathbb{Z})$$

We are missing a description of the cup product in terms of Hochster's formula. It is given by the following definition.

Definition 8.3.8 (Ring Structure in Hochster's Formula)

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define a ring structure in $\bigoplus_{J \subseteq \{1, \dots, n\}} \tilde{H}^*(\mathcal{K}_J; \mathbb{Z})$ as follows. Let α_L be the cochain basis element in $C^p(\mathcal{K}_I; \mathbb{Z})$ corresponding to the simplex $L \subseteq I$. The ring structure is induced by the map

$$C^p(\mathcal{K}_I; \mathbb{Z}) \otimes_{\mathbb{Z}} C^q(\mathcal{K}_J; \mathbb{Z}) \rightarrow C^{p+q+1}(\mathcal{K}_{I \cup J})$$

defined by

$$\alpha_L \otimes \alpha_M \mapsto \begin{cases} \varepsilon(L, I) \varepsilon(M, J) \varepsilon(L \cup M, I \cup J) \alpha_{L \sqcup M} & \text{if } I \cap J = \emptyset \\ 0 & \text{otherwise} \end{cases}$$

where $\varepsilon(L, I) = \prod_{j \in L} \varepsilon(j, I)$ and $\varepsilon(j, J) = (-1)^{r-1}$ if j is in the r th position in J .

Proposition 8.3.9

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- The ring structure in Hochster's formula is equal to the relative Kunneth theorem for

$$H^*(|\mathcal{K}_I * \mathcal{K}_J|) = H^*(\Sigma(|\mathcal{K}_I| \wedge |\mathcal{K}_J|)) = \tilde{H}^{*-1}(|K_I| \times |K_J|, |K_I \vee K_J|)$$

via the inclusion $\mathcal{K}_{I \sqcup J} \hookrightarrow \mathcal{K}_I * \mathcal{K}_J$.

- The natural isomorphism in the above prp induces a natural isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \bigoplus_{J \subseteq \{1, \dots, n\}} \tilde{H}^*(\mathcal{K}_J; \mathbb{Z})$$

While the cohomology groups can be easily computed using Hochster's formula, the ring structure can be seen more easily using the DGA.

Example 8.3.10

Let $\mathcal{K}$ be the standard polygon with vertices given by $\{1, \dots, 5\}$ and edges given by $\{\{1, 2\}, \dots, \{4, 5\}, \{1, 5\}\}$. We compute that

- $H^0(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = \bigoplus_{J \subseteq \{1, \dots, 5\}} \tilde{H}^{0-|J|-1}(\mathcal{K}_J; \mathbb{Z}) = \tilde{H}^{-1-|0|}(\mathcal{K}_{\emptyset}; \mathbb{Z}) = \mathbb{Z}$.
- $H^1(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = H^2(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = H^5(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = H^6(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) = 0$.

Also we have

$$\begin{aligned} H^3(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) &= \bigoplus_{J \subseteq \{1, \dots, 5\}} \tilde{H}^{2-|J|}(\mathcal{K}_J; \mathbb{Z}) \\ &= \tilde{H}^0(\mathcal{K}_{\{1,3\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{1,4\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{2,4\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{2,5\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{3,5\}}; \mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{3\}}] \oplus \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \\ &= \mathbb{Z}[u_1 v_3] \oplus \mathbb{Z}[u_1 v_4] \oplus \mathbb{Z}[u_2 v_4] \oplus \mathbb{Z}[u_2 v_5] \oplus \mathbb{Z}[u_3 v_5] \end{aligned}$$

$$\begin{aligned} H^4(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) &= \bigoplus_{J \subseteq \{1, \dots, 5\}} \tilde{H}^{3-|J|}(\mathcal{K}_J; \mathbb{Z}) \\ &= \tilde{H}^0(\mathcal{K}_{\{1,2,4\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{2,3,5\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{1,3,4\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{2,4,5\}}; \mathbb{Z}) \oplus \tilde{H}^0(\mathcal{K}_{\{1,3,5\}}; \mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \oplus \mathbb{Z}[\alpha_{\{1\}}] \oplus \mathbb{Z}[\alpha_{\{2\}}] \oplus \mathbb{Z}[\alpha_{\{3\}}] \\ &= \mathbb{Z}[-u_1 u_2 v_4] \oplus \mathbb{Z}[-u_2 u_3 v_5] \oplus \mathbb{Z}[-u_3 u_4 v_1] \oplus \mathbb{Z}[-u_4 u_5 v_2] \oplus \mathbb{Z}[u_1 u_5 v_3] \end{aligned}$$

and

$$\begin{aligned} H^7(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) &= \bigoplus_{J \subseteq \{1, \dots, 5\}} \tilde{H}^{6-|J|}(\mathcal{K}_J; \mathbb{Z}) \\ &= \tilde{H}^1(\mathcal{K}; \mathbb{Z}) \\ &= \mathbb{Z}[\alpha_{\{1,2\}}] \\ &= \mathbb{Z}[-u_3 u_4 u_5 v_1 v_2] \end{aligned}$$

where we identified the generators in Hochster's formula to generators in the DGA via the given formulas. Denote the generators of H^3 by $x_1, \dots, x_5$ and denote the generators of H^4 by $y_1, \dots, y_5$ and denote the generator of H^7 by z . Then we have the relations $x_1 y_4 = x_2 y_2 = -x_3 y_5 = x_4 y_3 = x_5 y_1 = z$, and 0 otherwise. Thus relabelling the generators give

$$H^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z}) \cong \frac{\Lambda_{\mathbb{Z}}[x_1, \dots, x_5, y_1, \dots, y_5]}{(x_i y_i - x_j y_j, x_i y_j, x_i x_j, y_i, y_j \mid 1 \leq i \neq j \leq 5)}$$

where $\deg(x_i) = 3$ and $\deg(y_i) = 4$. Notice that this is precisely the cohomology ring of $(S^3 \times S^4)^{\#5}$.

There are three descriptions of the cohomology groups:

$$H^k(\mathcal{Z}_K; \mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p} \left(\Lambda \left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}] u_i \right) \right)_{2q} = \bigoplus_{J \subseteq \{1, \dots, n\}} \tilde{H}^{k-|J|-1}(\mathcal{K}_J; \mathbb{Z})$$

For $L \subseteq J$ such that α_L is a cochain basis in the Hochster formula, we have

$$[\varepsilon(L, J)\chi(J \setminus L, L)] \leftrightarrow [\varepsilon(L, J)u_{J \setminus L}v_L] \leftrightarrow \alpha_L$$

Proposition 8.3.11

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- $\text{rank}(H^0(C^{-*,0}(\mathcal{Z}_K; \mathbb{Z}))) = 1$.
- $\text{rank}(H^0(C^{-*,2q}(\mathcal{Z}_K; \mathbb{Z}))) = 0$ for $q \neq 0$.
- $\text{rank}(H^{-p}(C^{-*,2q}(\mathcal{Z}_K; \mathbb{Z})))$ for $p > q$ or $q > n$.
- $\text{rank}(H^{-p}(C^{-*,2q}(\mathcal{Z}_K; \mathbb{Z})))$ for $p \geq q > 0$ or $q - p > \dim(\mathcal{K}) + 1$.
- $\text{rank}(H^{-1}(C^{-*,4}(\mathcal{Z}_K; \mathbb{Z}))) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- $\text{rank}(H^{-(n-\dim(\mathcal{K})+1)}(C^{-*,2n}(\mathcal{Z}_K; \mathbb{Z}))) = \text{rank}(\tilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Proposition 8.3.12

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- $\text{rank}(H^0(\mathcal{Z}_K; \mathbb{Z})) = 1$.
- $\text{rank}(H^1(\mathcal{Z}_K; \mathbb{Z})) = \text{rank}(H^2(\mathcal{Z}_K; \mathbb{Z})) = 0$.
- $\text{rank}(H^3(\mathcal{Z}_K; \mathbb{Z})) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- $\text{rank}(H^{n+\dim(\mathcal{K})+1}(\mathcal{Z}_K; \mathbb{Z})) = \text{rank}(\tilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Lemma 8.3.13

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1} , then the top cohomology $H^{n+m}(\mathcal{Z}_K; \mathbb{Z})$ is generated by $[\chi(\{1, \dots, n\} \setminus I, I)]$ for any $I \in \mathcal{K}$ and such that $|I| = n$.

If we use the Koszul complex as the cohomology model for $\mathcal{Z}_K$, then the top cohomology is generated by $u_{\{1, \dots, n\} \setminus I}v_I$ for any $I \in \mathcal{K}$ such that $|I| = n$.

9 The Real Moment Angle Complex

Recall that cubical complex of a simplicial complex. We know that

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

and

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} = \bigcup_{\sigma \in \mathcal{K}} \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

We consider the fat wedge filtration of $\mathbb{R}\mathcal{Z}_{\mathcal{K}}$. We note that at the i th step, we have

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}}^i = \bigcup_{I \subseteq \{1, \dots, n\}, |I|=i} \mathbb{R}\mathcal{Z}_{\mathcal{K}_I}$$

Proposition 9.0.1

Let $n \in \mathbb{N}^+$. Let $\mathcal{K}$ be a simplicial complex on n . Let $J \subseteq \mathcal{K}$. Then we have

$$(|\text{Cone}(\text{Bd}(\mathcal{K}_J))|, |\text{Bd}(\mathcal{K}_J)|) \cong (\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}, \mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1})$$

Proof

Let $i_c : |\Delta^n| \rightarrow I^{n+1}$ denote the cubical embedding. For each $J \subseteq \{1, \dots, n\}$, the embedding restricts to a map $i_c^J : |\mathcal{K}_J| \rightarrow I^J$ where $I^J = (I, 0)^J$ is the moment angle complex of I . For each $\mu \subseteq \tau \subseteq J$, denote $C_{\mu \subseteq \tau}^J$ the $C_{\mu \subseteq \tau}$ face of I^J (so $C_{\mu \subseteq \tau}^J$ still lives inside I^J but some coordinates are 0). We compute that

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1} = \bigcup_{A \subseteq J, |A|=|J|-1} \bigcup_{\sigma \in \mathcal{K}_A} \bigcup_{\substack{\mu \subseteq \tau \subseteq A \\ \tau \setminus \mu = \sigma}} C_{\sigma \subseteq \tau}^A = |\text{Bd}(\mathcal{K}_J)|$$

and similarly

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J} = \bigcup_{\sigma \in \mathcal{K}_J} \bigcup_{\substack{\mu \subseteq \tau \subseteq J \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}^J = |\text{Cone}(\text{Bd}(\mathcal{K}_J))|$$

■

Let us verify this for the case $\mathcal{K} = \partial\Delta^2$. Pick $I = \{1\}$. Let $i_c : |\mathcal{K}| \rightarrow I^n$ be the map that is the cubical embedding. Now the relative version of the identity

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

says that if $\sigma \in \mathcal{K}_I$, then

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}^I$$

where we consider $(D^1, S^0)^\sigma$ as a subspace of $\mathcal{Z}_{\mathcal{K}}$ by considering $\mathcal{Z}_{\mathcal{K}_I}$ as a subspace of $\mathcal{Z}_{\mathcal{K}}$, and on the right $C_{\mu \subseteq \tau}^I$ is the usual meaning but restricted to the cube $(D^1)^I$ of I^3 .

Restricting this map to $\mathcal{K}_I$, this is the map sending $\{1\}$ to $C_{\{1\}, \{1\}}^I$ which is the point $-1 \in C_{\{1\}, \{1\}}^I = D^1$, but is the point $(-1, 1, 1)$ in I^3 .

10 Algebra

10.1 Algebra, Coalgebra and Bialgebra

Let R be a commutative ring. Recall that an R -algebra is a ring A together with a ring homomorphism $R \rightarrow A$. Equivalently, it is an R -module with multiplication that is associative and respects the R -module structure.

Proposition 10.1.1

Let R be a commutative ring. Let $A \in \mathbf{Mod}_R$ be an R -module. Let $m : A \otimes_R A \rightarrow A$ and $u : R \rightarrow A$ be R -module homomorphisms. Then (A, m, u) is an associative algebra with multiplication given by m and identity given $u(1)$ if and only if the following diagrams commute:

$$\begin{array}{ccc} (A \otimes_R A) \otimes_R A & \xrightarrow{m \otimes 1} & A \otimes_R A \xrightarrow{m} A \\ \alpha \downarrow & & \uparrow m \\ A \otimes_R (A \otimes_R A) & \xrightarrow{1 \otimes m} & A \otimes_R A \end{array} \quad \begin{array}{ccccc} R \otimes_R A & \xrightarrow{u \otimes \text{id}_A} & A \otimes_R A & \xleftarrow{\text{id}_A \otimes u} & A \otimes R \\ & \searrow \lambda & \downarrow m & \swarrow \mu & \\ & & A & & \end{array}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Equivalently, the above proposition is the same as saying (A, m, u) is a monoid object in the monoidal category $(\mathbf{Mod}_R, \otimes_R, R)$. Motivated by the categorical description of an R -algebra, we define coalgebras as the formal dual of the above object in the following way.

Definition 10.1.2 (Coalgebras)

Let R be a commutative ring. Let $C \in \mathbf{Mod}_R$ be an R -module. Let $\Delta : C \rightarrow C \otimes_R C$ and $\varepsilon : C \rightarrow R$ be R -module homomorphisms. We say that (C, Δ, ε) is a coalgebra if the following diagrams commute:

$$\begin{array}{ccc} C & \xrightarrow{\Delta} & C \otimes_R C \xrightarrow{\Delta \otimes 1} (C \otimes_R C) \otimes_R C \\ & \searrow \Delta & \downarrow \alpha \\ & & C \otimes_R C \xrightarrow{1 \otimes \Delta} C \otimes_R (C \otimes_R C) \end{array} \quad \begin{array}{ccccc} & & C & & \\ & \swarrow \lambda & \downarrow \Delta & \searrow \mu & \\ R \otimes_R C & \xleftarrow{\varepsilon \otimes \text{id}_A} & C \otimes C & \xrightarrow{\text{id}_A \otimes \varepsilon} & C \otimes_R R \end{array}$$

where α is the natural isomorphism from associativity of tensor products, and λ, μ is the natural isomorphism from the unitality of tensor products.

Definition 10.1.3 (Bialgebras)

Let R be a commutative ring. Let $B \in \mathbf{Mod}_R$ be an R -module. Let $m : B \otimes_R B \rightarrow B, u : R \rightarrow B, \Delta : B \rightarrow B \otimes_R B$ and $\varepsilon : B \rightarrow R$ be R -module homomorphisms. We say that B is an R -bialgebra if the following are true.

- (B, m, u) is an R -algebra.
- (B, Δ, ε) is an R -coalgebra.
- Multiplication and comultiplication are compatible:

$$\begin{array}{ccccc} B \otimes_R B & \xrightarrow{m} & B & \xrightarrow{\Delta} & B \otimes_R B \\ \Delta \otimes \Delta \downarrow & & & & \uparrow m \otimes m \\ B \otimes_R B \otimes_R B \otimes_R B & \xrightarrow{\text{id} \otimes \tau \otimes \text{id}} & B \otimes_R B \otimes_R B \otimes_R B & & \end{array}$$

where τ is the natural isomorphism from commutativity of tensor products.

- Multiplication and counit are compatible:

$$\begin{array}{ccc} B \otimes_R B & \xrightarrow{m} & B \\ \varepsilon \otimes \varepsilon \downarrow & & \downarrow \varepsilon \\ R \otimes_R R & \xrightarrow{\lambda} & R \end{array}$$

where λ is the natural isomorphism from unitality of tensor products.

- Comultiplication and unit are compatible:

$$\begin{array}{ccc} R & \xrightarrow{u} & B \\ \lambda \downarrow & & \downarrow \Delta \\ R \otimes_R R & \xrightarrow{\Delta \otimes \Delta} & B \otimes_R B \end{array}$$

where λ is the natural isomorphism from unitality of tensor products.

- Unit and counit are compatible:

$$\begin{array}{ccc} R & \xrightarrow{\text{id}} & R \\ & \searrow u & \nearrow \varepsilon \\ & B & \end{array}$$

10.2 Hopf Algebra

Definition 10.2.1 (Hopf Algebras)

Let R be a commutative ring. Let $(H, m, u, \Delta, \varepsilon)$ be an R -bialgebra. Let $S : H \rightarrow H$ be an R -module homomorphism. We say that $(H, m, u, \Delta, \varepsilon, S)$ is a Hopf algebra if the following diagram commutes:

$$\begin{array}{ccccc} & & H \otimes_R H & \xrightarrow{S \otimes \text{id}_H} & H \otimes_R H \\ & \nearrow \Delta & & & \searrow m \\ H & \xrightarrow{\varepsilon} & R & \xrightarrow{u} & H \\ & \searrow \Delta & & & \nearrow m \\ & & H \otimes_R H & \xrightarrow{\text{id}_H \otimes S} & H \otimes_R H \end{array}$$

Definition 10.2.2 (Commutative and Cocommutative)

Let R be a commutative ring. Let H be a Hopf algebra.

- We say that H is commutative if the underlying algebra is commutative.
- We say that H is cocommutative if the underlying coalgebra is cocommutative.

Proposition 10.2.3

Let R be a commutative ring. Let H be a Hopf algebra. If H is a projective and finitely generated R -module, then $\text{Hom}_R(H, R)$ is a Hopf algebra. Moreover, the following are true.

- $\text{Hom}_R(M, R)$ is commutative if and only if H is cocommutative.
- $\text{Hom}_R(M, R)$ is cocommutative if and only if H is commutative.

10.3 Differential Graded Algebras

Definition 10.3.1 (Differential Graded Algebras)

Let R be a commutative ring. A differential graded R -algebra consists of the following data.

- A $\mathbb{Z}$ -graded R -algebra A .
- An R -algebra homomorphism of degree -1 (called homological dgas) or of degree 1 (called cohomological dgas).

such that the following are true.

- Differential: $d \circ d = 0$.
- Graded Leibniz: $d(ab) = (da)b + (-1)^{\deg(a)}a(db)$ for a, b homogeneous elements of A .

The differential property means that any (homological) differential graded algebra decomposes into a chain complex whose differential is given by d .

Definition 10.3.2 (Homomorphisms of Differential Graded Algebras)

Let R be a commutative ring. Let A, B be differential graded algebras. A morphism $\varphi : A \rightarrow B$ of differential graded algebras is a graded R -algebra homomorphism.

Lemma 10.3.3

Let R be a commutative ring. Let (A, d) be a cohomological differential graded algebra. Then $H^*(A, d) = \bigoplus_{n \in \mathbb{Z}} H^n(A, d)$ is a graded ring with multiplication defined by

$$[u] \cdot [v] = [uv]$$

for $u \in H^n(A, d)$ and $v \in H^m(A, d)$.

Proof

Notice that by the graded Leibniz rule, we have $d(uv) = (du)v + (-1)^n u d(v) = 0$ since u and v are cocycles. Hence uv is also a cocycle. ■

Products of DGA, tensor products, hom set

Definition 10.3.4 (Commutative Differential Graded Algebras)

Let R be a commutative ring. Let A be a differential graded algebra. We say that A is commutative (cdga) if A is graded commutative.

Lemma 10.3.5

Let R be a commutative ring. Let M be an R -module. Let $\varphi : M \rightarrow R$ be an R -module homomorphism. Then $K_\bullet(\varphi)$ is a commutative differential graded algebra.

Proposition 10.3.6

Let k be a field. Let A, B be cohomological differential graded algebras. Then the following are true.

- There is a natural isomorphism $H^*(A) \otimes_k H^*(B) \cong H^*(M \otimes_k N)$.
- There is a natural isomorphism $H^*(\text{Hom}_k(A, B)) \cong \text{Hom}_k(H^*(A), H^*(B))$

10.4 Models for DGAs

11 Koszul Algebras and Quadratic Algebras

12 Profinite Groups

12.1 Basic Properties

Definition 12.1.1 (Profinite Groups)

Let G be a topological group. We say that G is a profinite group if there exists an inverse system $X : \mathcal{I} \rightarrow {}_G\mathbf{Top}$ consisting of finite groups with the discrete topology such that

$$G = \varprojlim_{\mathcal{I}} X$$

In particular, this means that G is equipped with the weak topology.

Proposition 12.1.2

Let G be a topological group. Then G is profinite if and only if G is compact and totally disconnected.

Corollary 12.1.3

Let G be a profinite group. Then the following are true.

- G is Hausdorff.
- G is locally compact.

Proposition 12.1.4

Let G be a compact topological group. Then G/G_0 is a profinite group.

Proposition 12.1.5

Let G be a profinite group. Let $H \leq G$ be a subgroup. Then the following are true.

- If H is closed, then H is profinite.
- If H is closed and normal, then G/H is profinite.
- H is open if and only if H is closed and $[G : H] < \infty$.

12.2 Profinite Completion

13 Equivariant Algebraic Topology

13.1 G-Homotopy

Definition 13.1.1 (G-Homotopy) Let G be a topological group and let X, Y be G -spaces. Let $f, g : X \rightarrow Y$ be G -equivariant maps. A G -homotopy from f to g is a homotopy $H : X \times I \rightarrow Y$ from f to g such that for each $t \in I$, the map

$$H(-, t) : X \rightarrow Y$$

is G -equivariant map.

13.2 Equivariant CW Complexes

13.3

Definition 13.3.1 (The Homotopy Quotient (Borel Construction))

Let G be a topological group. Let X be a G -space. Define the homotopy quotient of X to be the orbit space

$$EG \times_G X = \frac{EG \times X}{G}$$

where the action of G on $EG \times X$ is given by $g \cdot (e, x) = (g \cdot e, g^{-1} \cdot x)$.

Proposition 13.3.2

Let G be a topological group. Let X be a G -space. Then

$$X \hookrightarrow EG \times_G X \rightarrow BG$$

is a fibration.

Lemma 13.3.3

Let G be a topological group. Let X be a G -space. If G acts freely on X , then there is a homotopy equivalence

$$EG \times_G X \simeq \frac{X}{G}$$

induced by the projection map $EG \times_G X \rightarrow X$.

14

Definition 14.0.1 (Gauge Group)

Let G be a topological group. Let $p : E \rightarrow X$ be a principal G -bundle. Define the gauge group of p to be the set

$$G(p) = \{f : E \rightarrow E \mid f \text{ is an isomorphism of } p\}$$

together with composition of functions.

Proposition 14.0.2

Let G be a topological group. Let $p : E \rightarrow X$ be a principal G -bundle. Then there is a group isomorphism

$$G(p) \cong \Gamma(\tilde{p})$$

where $\tilde{p}$ is the associated principal bundle where G acts on the fibers by the adjoint representation.

Note: The associated principal bundle is the space

$$\frac{E \times G}{\sim}$$

where $(p, g) \sim (ph^{-1}, hgh^{-1})$ for $p \in E$ and $g, h \in G$.

15

There is a one to one bijection between representations of $\pi_1(X)$ and locally constant sheaves (called local systems).

Definition 15.0.1

Let X be a space. Let $x_0 \in X$. Define the functor $\text{Fib}_{x_0} : \text{Cov}(X) \rightarrow \pi_1(X, x_0) \mathbf{Sets}$ as follows.

- For $p : \tilde{X} \rightarrow X$ a covering space, define $\text{Fib}_{x_0}(p) = p^{-1}(x_0)$.
- For $f : \tilde{X}_1 \rightarrow \tilde{X}_2$ a map of covering spaces, define $\text{Fib}_{x_0}(f)$ to be $f|_{p^{-1}(x_0)}$.

Theorem 15.0.2

Let X be a connected and locally simply connected space. Let $x_0 \in X$. Then

$$\text{Fib}_{x_0} : \text{Cov}(X) \rightarrow \pi_1(X, x_0) \mathbf{Sets}$$

is fully faithful and essentially surjective.

Theorem 15.0.3 (Classification of Local Systems)

Let X be a space (locally compact, Hausdorff, second countable, locally path connected, semi locally path connected, path connected). Then there is an equivalent of categories

$$\text{Func}(\mathbf{Open}^{\text{op}}, \mathbf{Vect}_{\mathbb{C}}) \simeq \text{Rep}(\pi_1(X))$$

Ref: Galois Groups and Fundamental Groups

Theorem 15.0.4 (Classification Theorem for Flat Connections)

Let M be a connected Riemannian manifold. Let G be a Lie group. Then there is a one to one correspondence

$$\frac{\{(P, A) \mid P \rightarrow M \text{ is a principal } G\text{-bundle and } A \text{ is a flat connection}\}}{\sim} \xleftrightarrow{1:1} \text{Rep}(\pi_1(X), G)$$

where $(P_1, A_1) \sim (P_2, A_2)$ if there exists bundle isomorphism $\varphi : P_1 \rightarrow P_2$ such that $\varphi^* A_2 = A_1$.

Ref: Differential Geometry; Taubes Bijection described in: Geometry of Characteristic classes; Morita

There is a natural bijection between principal bundles and vector bundles by taking the associated bundle. If $G = GL(V)$ for some finite dimensional vector space V over the field $k = \mathbb{R}$ or $\mathbb{C}$, the sheaf of sections of the associated bundle is a locally constant sheaf.

Also: Upgrade to diffeomorphism.

16 Infinite Galois Theory

Proposition 16.0.1

Let F be a field. Let K/F be a Galois extension. Then the following are true.

- The inverse limit of the system $\text{Gal}(L/F)$ for L/F a finite Galois extension is $\text{Gal}(K/F)$.
- $\text{Gal}(K/F)$ is a profinite group.
- The projection maps $\text{Gal}(K/F) \rightarrow \text{Gal}(L/F)$ for L/F a finite Galois extension is a surjective homomorphism.
- Let $F < L < K$ be a field extension. Then K/L is a Galois extension and $\text{Gal}(K/L)$ is a closed subgroup of $\text{Gal}(K/F)$.

Theorem 16.0.2 (Krull)

Let F be a field. Let K/F be a field extension. Then there is an inclusion reversing bijection

$$\left\{ \begin{array}{c} \text{Closed subgroups of} \\ \text{Aut}_F(K) \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{c} \text{Intermediary subfields} \\ F < L < K \end{array} \right\}$$

given as follows.

- For a subgroup $H \leq \text{Aut}_F(K)$, $H \mapsto \text{Fix}_F(H)$.
- For an intermediary subfield M , $M \mapsto \text{Aut}_M(K)$.

Proposition 16.0.3

Let F be a field. Let K/F be a Galois extension. Then the above bijection restricts to a bijection

$$\left\{ \begin{array}{c} \text{Normal Subgroups} \\ \text{of Aut}_K(L) \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{c} \text{Normal Extensions} \\ K < M \end{array} \right\}$$

17 Local Cohomology

17.1

Definition 17.1.1 (The I -Torsion of a Module)

Let R be a Noetherian commutative ring. Let M be an R -module. Let I be an ideal of R . Define the I -torsion of M to be

$$\Gamma_I(M) = \{m \in M \mid I^t m = 0 \text{ for some } t \in \mathbb{N}\}$$

Definition 17.1.2 (I -Torsion Modules)

Let R be a Noetherian commutative ring. Let M be an R -module. Let I be an ideal of R . We say that M is I -torsion if $M = \Gamma_I(M)$.

Definition 17.1.3 (The Torsion Functor)

Let R be a Noetherian commutative ring. Let I be an ideal of R . Define the I -torsion functor

$$\Gamma_I : {}_R\mathbf{Mod} \rightarrow {}_R\mathbf{Mod}$$

by the following.

- For each R -module M , the functor sends M to the I -torsion $\Gamma_I(M)$.
- For each R -module homomorphism $f : M \rightarrow N$, the functor sends f to $\Gamma_I(f) = f|_{\Gamma_I(M)}$.

Lemma 17.1.4

Let R be a Noetherian commutative ring. Let I be an ideal of R . Then Γ_I is a left exact functor.

Definition 17.1.5 (Local Cohomology)

Let R be a Noetherian commutative ring. Let M be an R -module. Let I be an ideal of R . Define the n th local cohomology of M with support in I to be

$$H_I^n(M) = (R^n \Gamma_I)(M)$$

Proposition 17.1.6

Let R be a Noetherian commutative ring. Let M be an R -module. Let I, J be ideals of R . If $\sqrt{I} = \sqrt{J}$, then the equality $\Gamma_I(M) = \Gamma_J(M)$ induces an equality

$$H_I^n(M) = H_J^n(M)$$

for all $n \in \mathbb{N}$.

Thus we only need to consider the local cohomology with support in radical ideals.

Lemma 17.1.7

Let R be a Noetherian commutative ring. Let M be an R -module. Let I be an ideal of R . Then $H_I^n(M)$ is I -torsion for all $n \in \mathbb{N}$.

Lemma 17.1.8

Let R be a Noetherian commutative ring. Let M_i be R -modules for each $i \in J$. Let I be an ideal of R . Then there is a natural isomorphism

$$H_I^n \left(\bigoplus_{i \in J} M_i \right) \cong \bigoplus_{i \in J} H_I^n(M_i)$$

Proposition 17.1.9

Let R be a Noetherian commutative ring. Let M be an R -module. Let I be an ideal of R . Then there are natural isomorphisms

$$\varinjlim_{i \in \mathbb{N}} \operatorname{Ext}_R^n(R/I^i, M) \cong H_I^n(M)$$

for all $n \in \mathbb{N}$.

18 Morita Theory

18.1

Definition 18.1.1 (Morita Equivalence)

Let R, S be rings. We say that R and S are Morita equivalent if there exists an equivalence of categories

$${}_R\mathbf{Mod} \simeq {}_S\mathbf{Mod}$$

Lemma 18.1.2

Let R, S be rings. Then R and S are Morita equivalent if and only if there exists an equivalence of categories

$$\mathbf{Mod}_R \simeq \mathbf{Mod}_S$$

Lemma 18.1.3

Let R, S be rings. Suppose that R and S are Morita equivalent. Let $F : {}_R\mathbf{Mod} \rightarrow {}_S\mathbf{Mod}$ be an equivalence. Then F is additive.

18.2 Progenerators

Definition 18.2.1 (Progenerators)

Let R be a ring. Let P be an R -module. We say that P is a progenerator of R if the following are true.

- P is finitely generated.
- P is projective.

Proposition 18.2.2

Let R, S be rings. Let $F : {}_R\mathbf{Mod} \rightarrow {}_S\mathbf{Mod}$ and $G : {}_S\mathbf{Mod} \rightarrow {}_R\mathbf{Mod}$ be additive functors. Then F and G are inverses if and only if there exists an (S, R) -bimodule P such that the following are true.

- ${}_S P$ is a progenerator.
- P_R is a progenerator.
- There are natural isomorphisms $F \cong P \otimes_R -$ and $G \cong \text{Hom}({}_S P, -)$.

Theorem 18.2.3 (Morita)

Let R, S be rings. Then the following are equivalent.

- R and S are Morita equivalent.
- There exists a progenerator R -module P_R such that $S \cong \text{End}_R(P_R)$.
- There exist an (R, S) -bimodule M and a (S, R) -bimodule N such that $M \otimes_S N \cong R$ as (R, R) -bimodules and $N \otimes_R M \cong S$ as (S, S) -bimodules.

18.3 Properties of Morita Equivalence

Proposition 18.3.1

Let R, S be rings. If R and S are Morita equivalent, then the following are true.

- There is a ring isomorphism $Z(R) \cong Z(S)$.
- $R/J(R)$ is Morita equivalent to $S/J(S)$.

Proposition 18.3.2

Let R, S be rings. Let M be an R -module. Suppose that R and S are Morita equivalent via the functor $F : {}_R\mathbf{Mod} \rightarrow {}_S\mathbf{Mod}$. Then the following are true.

- M is injective if and only if $F(M)$ is injective.
- M is projective if and only if $F(M)$ is projective.
- M is simple if and only if $F(M)$ is simple.

- M is semisimple if and only if $F(M)$ is semisimple.
- M is finitely generated if and only if $F(M)$ is finitely generated.
- M is Noetherian if and only if $F(M)$ is Noetherian.
- M is Artinian if and only if $F(M)$ is Artinian.

Proposition 18.3.3

Let R, S be rings. Suppose that R and S are Morita equivalent. Then the following are equivalent.

- R is simple if and only if S is simple.
- R is semisimple if and only if S is semisimple.
- R is Noetherian if and only if S is Noetherian.
- R is Artinian if and only if S is Artinian.

Proposition 18.3.4

Let R, S be rings. Then R and S are Morita equivalent if and only if ${}_R\mathbf{FGMod} \simeq {}_S\mathbf{FGMod}$.

19

Trick: If K is a triangulation of a sphere, then $K_{[n]\setminus J}$ is a deformation retract of $K \setminus K_J$. Now if $|K| \cong S^m$, (claim: K_J is a compact locally contractible subspace of $|K|$) then by Alexander duality we have

$$\tilde{H}^i(K_J) \cong \tilde{H}_{m-i-1}(S^m \setminus K_J) \cong \tilde{H}_{m-i-1}(K \setminus K_J)$$

Then we have

$$\tilde{H}^i(K_J) \cong \tilde{H}_{m-i-1}(K_{[n]\setminus J})$$

$$H^*(\mathcal{Z}_{\mathcal{K}_1}) \cong H^*(\mathcal{Z}_{\mathcal{K}_2}) \cong H^*(\mathcal{Z}_{\mathcal{K}_3}) \cong H^*((S^3 \times S^9)^{\#6} \# (S^4 \times S^8)^{\#8} \# (S^5 \times S^7)^{\#3})$$

$$H^*(\mathcal{Z}_{\mathcal{K}_4}) \cong H^*(\mathcal{Z}_{\mathcal{K}_5}) \cong H^*(\mathcal{Z}_{\mathcal{K}_6}) \cong H^*((S^3 \times S^9)^{\#5} \# (S^4 \times S^8)^{\#6} \# (S^5 \times S^7)^{\#4} \# (S^6 \times S^6)^{\#2})$$

$$H^*(\mathcal{Z}_{\mathcal{K}_7}) \cong H^*((S^3 \times S^9)^{\#3} \# (S^4 \times S^8)^{\#3} \# (S^5 \times S^7)) \otimes \frac{\mathbb{Z}\langle a, b, c, d, e, f \rangle}{(af - ce, bf - de, ad - bc, a^2, b^2, c^2, d^2, e^2, f^2)}$$

where $\deg(a) = \deg(b) = 3$, $\deg(c) = \deg(d) = 4$, $\deg(e) = 5$, $\deg(f) = 6$.

$$H^*(\mathcal{Z}_{\mathcal{K}_{34}}) \cong H^*(S^3 \times S^3 \times S^3 \times S^3)$$

20 Topics in Riemannian Geometry

21 Vector Bundles on Complex Manifolds

21.1 Serre's Correspondence

Theorem 21.1.1 (Serre's Correspondence)

Let M be a topological manifold. Then the following are true.

- There is an equivalence of categories

$$\mathbf{Vect}_n^{\mathbb{R}}(M) \simeq \mathbf{Loc}_n(M, \mathcal{C}_M^0)$$

- Suppose that M is a smooth manifold. Then there is an equivalence of categories

$$\mathbf{SmVect}_n^{\mathbb{R}}(M) \simeq \mathbf{Loc}_n(M, \mathcal{C}_M^{\infty})$$

- Suppose that M is a smooth manifold. Then there is an equivalence of categories

$$\mathbf{SmVect}_n^{\mathbb{C}}(M) \simeq \mathbf{Loc}_n(M, \mathcal{C}_M^{\infty} \otimes_{\mathbb{R}} \mathbb{C})$$

- Suppose that M is a complex manifold. Then there is an equivalence of categories

$$\mathbf{HolVect}_n^{\mathbb{C}}(M) \simeq \mathbf{Loc}_n(M, \mathcal{O}_M)$$

21.2 Holomorphic Line Bundles

By Serre's correspondence, holomorphic line bundles correspond to locally free sheaves of $\mathcal{O}_M$ -modules of rank 1, and recall isomorphism classes of these objects form a group structure with the tensor product, and it is group isomorphic to $H^1(M, \mathcal{O}_M^*)$.

Definition 21.2.1 (The Exponential Sequence)

Let M be a complex manifold. Define the exponential sequence of M to be the short exact sequence of sheaves of $\mathcal{O}_M$ -modules given by

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_M \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_M^* \longrightarrow 0$$

Proposition 21.2.2

Let M be a complex manifold. Let L be a holomorphic line bundle on M . Then the first chern class of L is given by the image of L in the homomorphism

$$\mathrm{Pic}(M) \cong H^1(M, \mathcal{O}_M^*) \rightarrow H^2(M, \mathbb{Z}) \cong H^2(M, \mathbb{Z})$$

induced by the long exact sequence of the exponential sequence.

21.3 Kodaira Dimension

Definition 21.3.1

Let M be a complex manifold. Let L be a holomorphic line bundle on M . Define ??? to be

$$R(M, L) = \bigoplus_{m=0}^{\infty} H^0(M, L^{\otimes m})$$

Definition 21.3.2 (The Canonical Ring)

Let M be a complex manifold. Define the canonical ring of M to be

$$R(M) = R(M, \mathcal{K}_M) = \bigoplus_{m=0}^{\infty} H^0(M, \mathcal{K}_M^{\otimes m})$$

Lemma 21.3.3

Let M be a complex manifold. If M is connected, then $R(M)$ is an integral domain.

Definition 21.3.4 (Kodaira Dimension)

Let M be a connected complex manifold. Define the Kodaira dimension of M to be

$$\text{kod}(M) = \begin{cases} -\infty & \text{if } R(X) = \mathbb{C} \\ \text{trdeg}_{\mathbb{C}}(Q(R(X))) - 1 & \text{otherwise} \end{cases}$$

21.4 Complex Analytic Varieties**Definition 21.4.1 (Complex Analytic Varieties)**

Let $Z \subseteq \mathbb{C}^n$. We say that Z is a complex analytic variety if there exists an open set $Z \subseteq U \subseteq \mathbb{C}^n$ and $f_1, \dots, f_k \in \mathcal{O}_{\mathbb{C}^n}(U)$ such that

$$Z = \mathbb{V}(f_1, \dots, f_k)$$

We use a subset of $\mathbb{C}^n$ in the definition because in $\mathbb{C}^n$, locally defined functions cannot in general be defined globally (compare it to the real case, where they have partitions of unity and bump functions).

Indeed, for smooth functions on $\mathbb{R}^n$, a function on an open set U can be modified (via bump functions) to produce a global function that agrees with it on a smaller open subset. You can always find a global function that captures the local behavior near any point.

For holomorphic functions on $\mathbb{C}^n$: no such modification is possible. A holomorphic function cannot be cut off or modified while remaining holomorphic, so local holomorphic data is genuinely trapped in its domain.

Definition 21.4.2 (The Structure Sheaf) Let $U \subseteq \mathbb{C}^n$ be an open subset. Let $f_1, \dots, f_k \in \mathcal{A}_{\mathbb{C}^n}(U)$ be analytic functions on U . Let $Z = \mathbb{V}(f_1, \dots, f_k)$ be the vanishing locus of $f_1, \dots, f_k$. Define the structure sheaf

$$\mathcal{O}_Z : \text{Open}(Z) \rightarrow \mathbf{CRing}$$

as follows.

- For each open set $U \subseteq Z$, define

$$\mathcal{O}_Z(U) = \frac{\mathcal{O}_{\mathbb{C}^n}(U)}{(f_1, \dots, f_k)}$$

- For each inclusion $U \subseteq V \subseteq Z$ of open sets, define

$$\mathcal{O}_Z(V) \rightarrow \mathcal{O}_Z(U)$$

to be the induced map of the restriction $\mathcal{O}_{\mathbb{C}^n}(V) \rightarrow \mathcal{O}_{\mathbb{C}^n}(U)$.

21.5 Complex Analytic Subvarieties of Manifolds**Definition 21.5.1 (Complex Analytic Subvarieties)**

Let M be a complex manifold. Let $Z \subseteq M$ be a subset. We say that Z is a complex analytic subvariety of M if for each $p \in Z$, there exists a chart (U, ϕ) of M such that $\phi(Z \cap U)$ is a complex analytic variety.

In particular, complex analytic subvarieties of $\mathbb{C}^n$ are just complex analytic varieties.

Theorem 21.5.2 (Cartan's (Coherence) Theorem) Let M be a complex manifold and let A be an analytic subset of M . Then $\mathcal{I}_A$ is a coherent sheaf of $\mathcal{O}_M$ -modules.

22 Complex Analytic Spaces

22.1 Complex Analytic Spaces

Definition 22.1.1 (Complex Analytic Spaces) Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is a complex analytic space if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is a complex analytic space.

Lemma 22.1.2

Let $(X, \mathcal{O}_X)$ be a complex analytic space. Then X is a complex manifold if and only if for each $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is an isomorphism

$$(U, \mathcal{O}_X|_U) \cong (V, \mathcal{O}_{\mathbb{C}^n}|_V)$$

for some open subset $V \subseteq \mathbb{C}^n$.

Theorem 22.1.3 Let X be a complex space. Then X_{reg} is dense and open in X . Moreover, X_{reg} consists of a disjoint union of complex manifolds.

Theorem 22.1.4 Let X be an irreducible complex space. Then every non-constant holomorphic function $f : X \rightarrow \mathbb{C}$ is an open map.

Corollary 22.1.5 Let X be an irreducible compact complex space. Then every holomorphic function $f : X \rightarrow \mathbb{C}$ on X is constant.

22.2 Complex Analytic Subvarieties

Definition 22.2.1 (Complex Analytic Subvarieties)

Let $(X, \mathcal{O}_X)$ be a complex analytic space. Let $Z \subseteq X$ be subset. We say that Z is a complex analytic subvariety if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ and $f_1, \dots, f_k \in \mathcal{O}_X(U)$ such that

$$Z \cap U = \mathbb{V}_U(f_1, \dots, f_k)$$

This definition generalizes complex analytic subvarieties of complex manifolds.

23 Analytic Manifolds

23.1 Real Analytic Manifolds

Definition 23.1.1 (Real Analytic Manifolds) A real analytic manifold is a topological manifold with analytic transition maps.

In the real case, every analytic manifold is a smooth manifold because every analytic function is necessarily infinitely differentiable. While not every analytic function is smooth, we can still equip a smooth structure on analytic manifolds.

Theorem 23.1.2 ((Grauert-Whitney)) Every smooth manifold admits a compatible real analytic structure.

Definition 23.1.3 (Analytic Functions) Let M be a real analytic manifold. Let $U \subseteq M$ be a subset. Let $f : U \rightarrow \mathbb{R}$ be a function. We say that f is analytic if for all $x \in U$, there exists a chart $(V, \varphi = x^1, \dots, x^n)$ of M such that

$$f \circ \varphi^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}$$

is analytic in the usual sense.

Definition 23.1.4 (Sheaf of Analytic Functions) Let M be an analytic manifold. Define the sheaf of analytic functions

$$\mathcal{A}_M : \mathbf{Open} \rightarrow \mathbf{CRing}$$

as follows.

- For each open set $U \subseteq M$, $\mathcal{A}_M(U)$ consists of the ring of analytic functions on U
- For each inclusion $V \subseteq U$, there is a unique morphism

$$\text{res}_V^U : \mathcal{A}_M(U) \rightarrow \mathcal{A}_M(V)$$

given by $f \mapsto f|_V$

24 Analytic Varieties

24.1 Real Analytic Varieties

24.2 Regular and Singular Points

Definition 24.2.1 (Regular and Singular Points) Let M be a complex manifold and let A be an analytic subset of M . We say that $x \in A$ is a regular point if there exists some open neighbourhood U of x such that $A \cap U$ is a complex submanifold of M . Otherwise, x is said to be singular. Denote

$$A_{\text{reg}} = \{x \in A \mid x \text{ is a regular point}\} \quad \text{and} \quad A_{\text{sing}} = \{x \in A \mid x \text{ is a singular point}\}$$

Theorem 24.2.2 Let M be a complex manifold and let A be an analytic subset of M . Then A_{sing} is an analytic subset of A .

25 Transcendental Algebraic Geometry

25.1

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

Definition 25.1.1 (The Analytification of a Scheme)

Let X be a scheme locally of finite type over $\mathbb{C}$. Define the analytification of X to be the unique analytic space X^{an} together with a morphism of locally ringed space $\alpha : X^{\text{an}} \rightarrow X$ satisfying the following universal property: For any analytic space Y and any morphism of locally ringed spaces $f : Y \rightarrow X$, there exists a unique morphism of analytic spaces $g : Y \rightarrow X^{\text{an}}$ such that the following diagram commutes:

$$\begin{array}{ccc} & Y & \\ \exists! g \swarrow & \downarrow f & \\ X^{\text{an}} & \xrightarrow{\alpha} & X \end{array}$$

Recall that the Yoneda embedding $h : \mathbf{LRS} \rightarrow \mathbf{Set}^{\mathbf{LRS}}$ is the functor sending a locally ringed space X to the functor $h_X : \mathbf{LRS} \rightarrow \mathbf{Set}$ defined by $h_X(Y) = \text{Hom}_{\mathbf{LRS}}(Y, X)$. Then the universal property of the analytification means that the functor h_X restricted to analytic spaces $\mathbf{An}$ is representable.

One thinks of the morphism $X^{\text{an}} \rightarrow X$ as doing two things. It is an enlargement of the standard topology of X^{an} to the Zariski topology of X , and it introduces generic points outside of $X(\mathbb{C})$ (Indeed we will see that X^{an} has the same underlying set as $X(\mathbb{C})$).

Theorem 25.1.2

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the analytification of X exists and is unique up to unique isomorphism.

Definition 25.1.3 (The Analytification Functor)

Define the analytification functor

$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \rightarrow \mathbf{An}$$

as follows.

- For each scheme X locally of finite type over $\mathbb{C}$, we have X^{an} is the analytification of X .
- For a morphism of schemes $f : X \rightarrow Y$ over $\mathbb{C}$, the universal property of Y^{an} induces a morphism $X^{\text{an}} \rightarrow Y^{\text{an}}$. Precomposing with the canonical morphism $X^{\text{an}} \rightarrow X^{\text{an}}$ gives the desired map $X^{\text{an}} \rightarrow Y^{\text{an}}$.

Proposition 25.1.4

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the underlying set of X^{an} is given by $X(\mathbb{C})$.

Proposition 25.1.5

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the following are true.

- X is separated if and only if X^{an} is Hausdorff.
- X is connected if and only if X^{an} is connected.
- X is smooth if and only if X^{an} is a complex manifold.
- X is proper if and only if X^{an} is compact.

Proposition 25.1.6

Let X be a variety over $\mathbb{C}$. Then the following are true.

- If X is irreducible, then X^{an} is connected.
- If X is smooth and affine, then X^{an} is a Stein manifold.
- If X is smooth and projective, then X^{an} is Kahler.

Theorem 25.1.7 (GAGA)

Let X be a proper scheme of finite type over $\mathbb{C}$. Then the following are true.

- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Then $\mathcal{F}^{\text{an}}$ is a sheaf of $\mathcal{O}_{X^{\text{an}}}$ -module.
- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. There is a natural isomorphism

$$H^i(X, \mathcal{F}) \cong H^i(X^{\text{an}}, \mathcal{F}^{\text{an}})$$

- The analytification functor induces an equivalence of categories

$$\mathbf{Coh}_{\mathcal{O}_X} \cong \mathbf{Coh}_{\mathcal{O}_{X^{\text{an}}}}$$

- Let $Y \subseteq X^{\text{an}}$ be a closed complex analytic subspace. Then there exists a closed subvariety Z of X such that $Z^{\text{an}} = Y$.

Corollary 25.1.8 (Chow's Theorem)

Let X be a closed complex analytic subspace of $\mathbb{CP}^n$. Then X is a projective variety over $\mathbb{C}$.

Theorem 25.1.9 (Grothendieck)

Let X be a smooth variety over $\mathbb{C}$. Then there is a natural isomorphism

$$H_{\text{dR}}^n(X) \cong H_{\text{Holo}}^n(X^{\text{an}}) \cong H_{\text{dR}}^n(X^{\text{an}})$$

induced by the canonical map $\alpha : X^{\text{an}} \rightarrow X$.

Corollary 25.1.10

Let X be a smooth variety over $\mathbb{C}$. Then the above isomorphism and de Rham's theorem induces a natural isomorphism

$$H_{\text{dR}}^n(X) \cong H_{\text{sing}}^n(X^{\text{an}}; \mathbb{C})$$

Theorem 25.1.11 (The Period Isomorphism)

Let k be a subfield of $\mathbb{C}$. Let X be a smooth variety over k . Then there is a natural isomorphism

$$H_{\text{dR}}^*(X) \otimes_k \mathbb{C} \cong H_{\text{sing}}^*(X^{\text{an}}; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

Proposition 25.1.12

Let X be a smooth projective variety over $\mathbb{C}$. Let $\mathcal{L}$ be a line bundle over X . Then $\mathcal{L}$ is ample if and only if $\mathcal{L}^{\text{an}}$ is positive.

Theorem 25.1.13 (Kodaira Embedding)

Let X be a compact complex manifold. Then the following are equivalent.

- There exists a projective variety Z over $\mathbb{C}$ such that $Z^{\text{an}} = X$.
- X admits a positive line bundle.
- X is Kahler and $H^{1,1}(X, \mathbb{Z}) \neq 0$.

25.2**Proposition 25.2.1**

Let X be a smooth projective curve over $\mathbb{C}$. Then X^{an} is homeomorphic to $\Sigma_{p_g(X)}$ the orientable surface of genus $p_g(X)$.

Corollary 25.2.2

There is an equivalence of categories between smooth projective curves over $\mathbb{C}$ and compact Riemann surfaces.

26 The Geometry of Fiber Bundles

Proposition 26.0.1 (Poincaré-Hopf Theorem)

Let M be a smooth manifold. Then we have

$$e(TM)([M]) = \chi(M)$$

26.1 The Normal Bundle

Definition 26.1.1 (The Normal Bundle)

Let M, N be smooth manifolds. Let $\iota : N \hookrightarrow M$ be a smooth embedding. Define the normal bundle of N in M to be the quotient bundle

$$N_{M/N} = \frac{TM}{TN}$$

Example 26.1.2

Let $\iota : S^n \rightarrow \mathbb{R}^{n+1}$ be the standard inclusion of the unit sphere. Then the following are true.

- $N_{\mathbb{R}^{n+1}/S^n} \cong \varepsilon$ is a trivial bundle of rank 1.
- $w_i(TS^n) = 1$ for all i .

Proof

Notice that $N_{\mathbb{R}^{n+1}/S^n}$ is a bundle of rank equal to 1. It is the trivial bundle if and only if it admits a nowhere vanishing section. Define a section $s : S^n \rightarrow N_{\mathbb{R}^{n+1}/S^n}$ by $s(x) = (x, x)$ where $x \in N_{\mathbb{R}^{n+1}/S^n}$ is a vector in the direction x . Then clearly it is non-vanishing.

Now we have $TS^n \oplus N_{\mathbb{R}^{n+1}/S^n} = T\mathbb{R}^{n+1}$. Applying the total Stiefel-Whitney class gives $w(TS^n) \cdot 1 = w(TS^n)w(N_{\mathbb{R}^{n+1}/S^n}) = w(T\mathbb{R}^{n+1}) = 1$. Hence $w(TS^n) = 1$. Since $w_0(TS^n) = 1$, the grading in the cohomology ring implies that $w_i(TS^n) = 0$ for all i . ■

26.2 Principle Bundles over Lie Groups

Proposition 26.2.1

Let M be a smooth manifold. Let G be a Lie group with a free and smooth action on M . Then the following are true.

- For each $x \in M/G$, there exists a neighbourhood U of x and a map $s : U \rightarrow M$ such that $\pi \circ s = \text{id}_U$, where $\pi : M \rightarrow M/G$ is the projection map.
- The projection map $M \rightarrow M/G$ is a principal G -bundle.

26.3 Connections on a Principal Bundle

Definition 26.3.1 (The Vertical Bundle)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Define the vertical bundle of P to be the smooth subbundle

$$VP = \coprod_{p \in P} \ker(d\pi_p)$$

Lemma 26.3.2

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let $p \in P$. Then there is an isomorphism

$$\iota_p : \text{Lie}(G) \xrightarrow{\cong} (VP)_p$$

given by $g \mapsto \left. \frac{de^{tg} \cdot p}{dt} \right|_{t=0}$.

Definition 26.3.3 (Horizontal Distributions)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . A horizontal distribution on P is a subbundle $HP \subseteq TP$ such that the following are true.

- $TP = VP \oplus HP$.
- $(HP)_{g \cdot p} = (L_g)_*(HP)_p$ for all $p \in P$ and $g \in G$.

Generalizes connections on vector bundles in the following way: For $p : E \rightarrow M$ a smooth vector bundle, a connection on $\text{Fr}(E)$ as a principal $\text{GL}(n, F)$ bundle is the same data as a connection on $p : E \rightarrow M$ as a vector bundle.

Definition 26.3.4 (Connection Forms on Principal Bundles)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . A connection 1-form on P is a smooth section $\omega \in \mathcal{C}^\infty(T^*P \otimes \text{Lie}(G))$ such that the following are true.

- We have $\omega_p(\iota_p) = \text{id}_{\text{Lie}(G)}$ for all $p \in P$.
- We have

$$\omega_{g \cdot p}((dL_g)_p(v)) = \text{Ad}(g)(\omega_p(v))$$

for all $g \in G, p \in P$ and $v \in T_p P$.

Proposition 26.3.5

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Then the following are true.

- Suppose that HP is a horizontal distribution on P . Define a 1-form ω pointwise by $\omega_p(X) = \iota_p^{-1}(X^V)$ where X^V is the vertical component of X in the decomposition $T_p P = (VP)_p \oplus (HP)_p$. Then ω is a connection 1-form.
- Suppose that ω is a connection 1-form on P . Then $\coprod_{p \in P} \ker(\omega_p)$ is a horizontal distribution on P .

Definition 26.3.6 (The Curvature Form of a Connection)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let ω be a connection 1-form. Define the curvature form of ω to be the 2-form $\Omega \in \mathcal{C}^\infty(\Lambda^2 T^*P \otimes \text{Lie}(G))$ given by

$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega]$$

This definition is motivated by the local description of the curvature form for vector bundles.

Using tensor algebra, we see that $T^*P \otimes T^*P \otimes \mathfrak{g} \cong TP \times TP \rightarrow (P \times \mathfrak{g})$. Since $\Lambda^2 T^*P$ is the subbundle of $T^*P \otimes T^*P$ consisting of alternating forms, we may also think of $\Omega \in \mathcal{C}^\infty(\Lambda^2 T^*P \otimes \mathfrak{g})$ as a map $TP \times TP \rightarrow P \times \mathfrak{g}$ satisfying $\Omega(X, X) = 0$. Also, for each $p \in P$, $\Omega_p : T_p P \times T_p P \rightarrow \mathfrak{g}$ is then an alternating bilinear map.

Definition 26.3.7 (Components of the Curvature Form)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let ω be a connection 1-form. Let Ω be the curvature form. Let $e_1, \dots, e_n$ be a basis of $\mathfrak{g}$. Define the components of Ω to be the 2-forms $\Omega^i \in \Omega^2(P)$ such that

$$\Omega = \sum_{i=1}^n \Omega^i \otimes e_i$$

26.4 The Chern-Weil Homomorphism

Let V be a finite dimensional vector space over a field k . Recall from Rings and Modules that there is an isomorphism of graded F -algebras $S(V) \cong k[x_1, \dots, x_n]$.

Definition 26.4.1 (Ad-Invariant Polynomials)

Let G be a Lie group. Let $f \in S(\text{Lie}(G)^*)$. We say that f is ad-invariant if $f(\text{Ad}_g(X)) = f(X)$ for all $X \in \text{Lie}(G)$.

Definition 26.4.2 (The Algebra of Ad-Invariant Polynomials)

Let G be a Lie group. Define $\mathbb{R}$ -algebra of ad-invariant polynomials of G by

$$I(G) = \{f \in S(\text{Lie}(G)^*) \mid f \text{ is ad-invariant} \}$$

Let $f \in S(\text{Lie}(G)^*)$ be a polynomial. Let Ω be a curvature 2-form. For each $p \in P$, we know that Ω_p is an alternating bilinear map from $T_p P$ to $\text{Lie}(G)$.

Definition 26.4.3 (Evaluating Polynomials on the Curvature)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let ω be a connection 1-form. Let $f \in I(G)$ be an ad-invariant polynomial of the form $f = \sum_{i_1, \dots, i_n} \alpha_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n}$. Let Ω be the curvature 2-form of ω . Define the evaluation of f on Ω to be the $2k$ -form $f(\Omega) \in \Omega^{2k}(M)$ given by

$$f(\Omega) = \sum_{i_1, \dots, i_n} \alpha_{i_1, \dots, i_n} (\Omega^1)^{\wedge i_1} \wedge \cdots \wedge (\Omega^n)^{\wedge i_n}$$

Lemma 26.4.4

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let $f \in I(G)$ be an ad-invariant polynomial. Let ω be a connection 1-form. Let Ω be the curvature form. Then we have

$$df(\Omega) = 0$$

Lemma 26.4.5

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let $f \in I(G)$ be an ad-invariant polynomial. Let ω, ω' be connection 1-forms. Let Ω and Ω' be the curvature forms respectively. Then we have

$$[f(\Omega)] = [f(\Omega')] \in H_{\text{dR}}^{2k}(M)$$

Definition 26.4.6 (The Chern-Weil Homomorphism)

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let Ω be the curvature form of any connection. Define the Chern-Weil homomorphism to be the map

$$P(\Omega) : S(\text{Lie}(G)^*) \rightarrow H_{\text{dR}}^*(M)$$

given by $f \mapsto [f(\Omega)]$.

Lemma 26.4.7

Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \rightarrow M$ be a smooth principal bundle over G . Let Ω be the curvature form of any connection. Then $P(\Omega)$ is an algebra homomorphism.

Proposition 26.4.8

Let M be a smooth manifold. Let $\pi : P \rightarrow M$ be a smooth vector bundle of dimension n over $\mathbb{C}$. Let Ω be the curvature form of any connection. Suppose that $f_0, \dots, f_n \in S(\mathfrak{gl}(n, \mathbb{C})^*)$ are such that $c_X(\lambda) = \sum_{i=0}^n f_i(X) \lambda^{n-i}$. Then the following are true.

- $f_0, \dots, f_n$ generate $S(\mathfrak{gl}(n, \mathbb{C})^*)$ as a $\mathbb{C}$ -algebra.

- $[f_1(\Omega)], \dots, [f_n(\Omega)]$ are the Chern classes of P .

Proposition 26.4.9

Let M be a smooth manifold. Let $\pi : P \rightarrow M$ be a smooth vector bundle of dimension n over $\mathbb{R}$. Let Ω be the curvature form of any connection. Suppose that $f_0, \dots, f_n \in S(\mathfrak{gl}(n, \mathbb{R})^*)$ are such that $c_X(\lambda) = \sum_{i=0}^n f_i(X) \lambda^{n-i}$. Then the following are true.

- $f_0, \dots, f_n$ generate $S(\mathfrak{gl}(n, \mathbb{R})^*)$ as a $\mathbb{R}$ -algebra.
- $[f_1(\Omega)], \dots, [f_n(\Omega)]$ are the Pontrjagin classes of P .